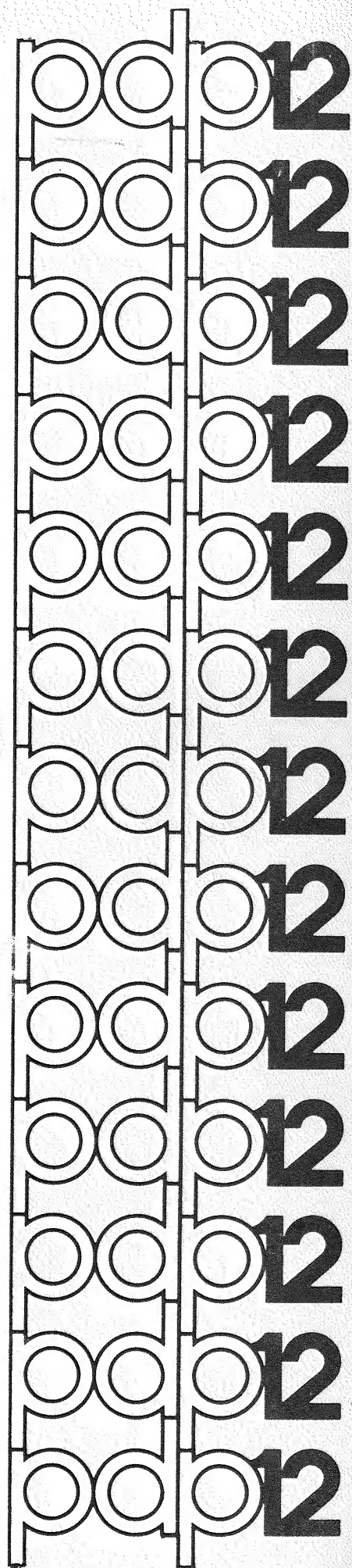
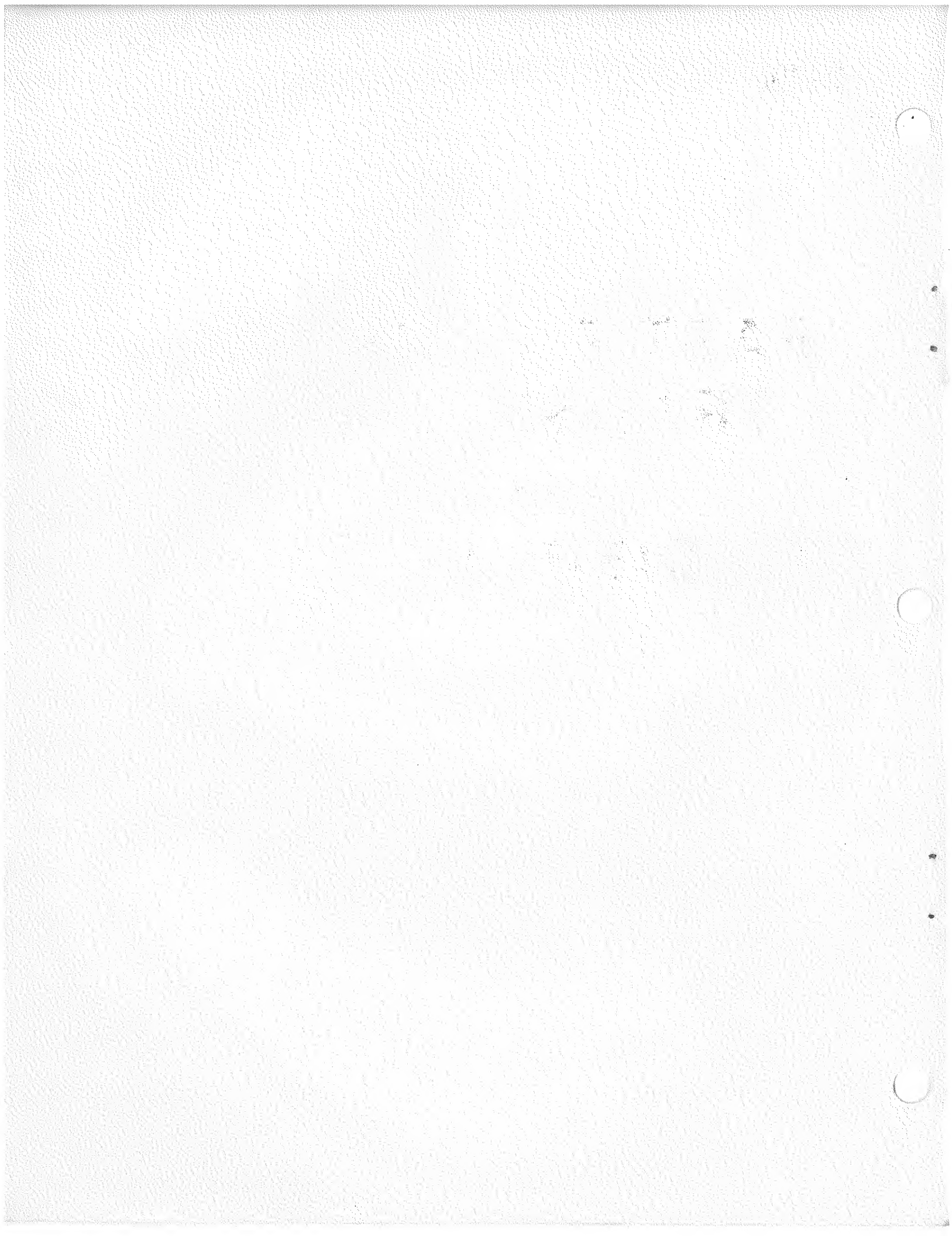


digital

FAST FOURIER TRANSFORM AND DISPLAY





FAST FOURIER TRANSFORM

AND

DISPLAY

PROGRAMMER'S REFERENCE MANUAL

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ACKNOWLEDGMENT

The PDP-12 Fast Fourier Transform + Display program is an adaptation of a program written by James Rothman, of Digital Equipment Corporation. The algorithm is described briefly in Section 7.0 of this manual and in detail in DECUSCOPE, Volume 7, Number 3, available from DECUS Library, Digital Equipment Corporation, Maynard, Massachusetts.

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1.0 INTRODUCTION

The FFTD (Fast Fourier Transform + Display) program can perform a Fast Fourier Transform or Inverse Fast Fourier Transform on 4 to 1024 real or complex points which have been stored on a LAP6-DIAL¹ or data LINC-tape or disk. The real and imaginary parts of the input or output data and the magnitude of the output data may be displayed on the scope via a moving window. Transformed data may also be stored on a DIAL or data LINCtape or disk. In addition, the scale of the displayed data can be user-modified over twelve different ranges.

2.0 MINIMUM HARDWARE REQUIREMENTS

8K PDP-12B with EAE. !

3.0 OPERATING PROCEDURE

3.1 Loading FFTD

FFTD is a "load and go" program and is called from tape or disk by the DIAL command:

→LO FFTD, n)

where n is the tape (0-7) or disk (10-17) containing the program. A DIAL system tape must be on unit 0. (If a non-existent unit is addressed, NO is displayed on the scope. Press RETURN and issue the proper command.)

At any time during program operation, FFTD may be restarted by pressing the console keys: LINC mode, I/O PRESET, and START 20.

3.2 FFTD Displays

The first display is:

```
DISPLAY 1          SINGLE PRECISION FFT
                   INPUT ON DIAL UNIT? Y/N__
```

¹LAP6-DIAL is hereafter referred to as DIAL.

Type Y if the data file is on a tape or disk containing DIAL; type N if the file is on a data tape or disk. (A file copied from paper tape via PIP must be referenced as a data tape or disk.)

The final user replies to all the scope displays are terminated by pressing LINE FEED.

If the input is on a DIAL tape or disk, the second display is:

```
DISPLAY 2          UNIT NUMBER__  
                   FILE NAME_____
```

Specify the unit number, 0 to 7 for tape, and 10 to 17 for disk, where the file is located and press RETURN. Then type the file name, which may be 1 to 3 characters long and must begin with a non-numeric character and not contain a ?, /, \, or >. After typing the file name, press LINE FEED. Note that a file addressed by name on a DIAL tape or disk can not have a header block and must have been placed on the device only by the FFTD program. If a non-existent unit is requested, NO is displayed. To restart the program from LINctape, press STOP, I/O PRESET, and START 20. The program must be reloaded from an RK8 or RF08 disk.

The user is told if the file is not on the specified unit:

```
DISPLAY 3          CANNOT FIND  
                   HIT RETURN TO CONT
```

Press RETURN to bring back display 2

If the input is on a data tape or disk, the second display is:

```
DISPLAY 4          UNIT NUMBER__  
                   BLOCK NUMBER_---
```

The unit may be any number from 0 to 7 for tape and 10 to 17 for disk. The block number must be an octal number from 0 to 777. If a data file with a header block is on a DIAL device, it may be accessed by this sequence (instead of the DIAL message). The correct block number is the value in the DIAL index plus one. After the file has been located, the calculation must be specified.

DISPLAY 5 HOW MANY PTS_____
 (4-1024 BY POWERS OF 2)
 REAL OR
 COMPLEX? R/C_

Powers of 2, from 2 to 10, are acceptable, permitting 4 to 1024 points. Type R if the data is real; type C if it is complex. (Refer to Section 4.0 for a description of data storage format.) If there is not enough room between the starting block number and the end of tape to hold the number of points specified, display 5 will reappear.

The calculation is further specified:

DISPLAY 6 FFT OR DISPLAY? F/D_
 TRANSFORM OR
 INVERSE? T/I_

If the data is just to be displayed, type D and press RETURN. Then type T if the data has most recently been transformed or I if it has not been manipulated at all or has been inversely transformed. Continue at display 7.

The next display is:

DISPLAY 7 OUTPUT ON DIAL UNIT? Y/N_

Type Y if output is to a DIAL tape or disk; type N if output is to a data tape or disk.

A reply of Y to display 7 (DIAL tape or disk) causes the display:

DISPLAY 8 UNIT NUMBER__
 FILE NAME_____

These answers have the same restrictions as the input display, display 2. If there is not enough space on the DIAL tape/disk to hold the output data, the next display is:

DISPLAY 9 NO SPACE
 HIT RETURN TO CONT

Press RETURN to bring back display 7.

If a file already exists with the specified name, the next display is:

DISPLAY 10 REPLACE? Y/N_

Type Y or N to replace or not to replace the file. A reply of N will cause display 8 to reappear. If the file is to be replaced, but the new file is larger than the old file, display 9 will reappear.

If output is to a data tape or disk, the next display is:

DISPLAY 11 UNIT NUMBER __
 BLK NUMBER ___

The answers have the same restrictions as the input display, display 4. If there is not enough space from the starting block number to the end of the tape to hold the output data, display 9 will reappear.

The program will now read in the data, perform a Fast Fourier Transform or Inverse Fast Fourier Transform, and write the results as complex data pairs onto the specified tape or disk.

When the transform is completed or if just displays are desired, the following message is displayed:

DISPLAY 12 WHICH DISPLAY?
 R(EAL)
 I(MAGINARY)
 M(MAGNITUDE)
 S(SCALE FACTOR)
 LINE FEED (RESTART)

Type R, I, M, or S and LINE FEED to obtain the desired display. The scale factor is displayed as a decimal number (0-12). (Refer to Section 6.0, Data Scaling, for an explanation of the scale factor.) (The magnitude, M, for $a+ib$ is $M = \sqrt{a^2+b^2}$.)

If the display is less than 512 points, it will be stationary and centered on the scope. If it contains 512 or more points, the display can be moved in either direction using A/D knob 0.

A cursor which can be moved by rotating A/D knob 1 will ride along the curve. Associated with the cursor are four octal words displayed in the top left corner of the scope, one beneath the other. The first two words are the absolute 15-bit core address of the cursor point. The third word is the contents of the displayed core address, i.e., the actual 12-bit value in the data buffer of the data word that corresponds

to the cursor point. The fourth word is the scope Y coordinate of the cursor point. The fourth word is a relative value and depends upon the Y scale factor and Y offset. Because the data is scaled to nine bits prior to display, the fourth word or Y coordinate will range from 0001 to 1000₈, where 0001 corresponds to the bottom of the scope and 1000 to the top.

The curve can be expanded in the Y direction by typing a 1 or decreased by typing Q. Twelve different ranges are possible. As the display is enlarged, no check is made against losing significant digits of large values because the user may wish to expand small features of the display. Therefore, as the display is enlarged, large values may suddenly decrease in size as significant digits are lost.

The magnitude display is shown at half scale initially. If the values allow, the number 1 can be typed once to show the display at full scale.

Pressing RETURN will cause display 12 to reappear. As many displays as desired may be requested. Subsequent displays will be initially shown at the same range as the preceding display. Pressing LINE FEED without entering a character will cause display 1 to reappear.

4.0 EXAMPLE

This section provides examples of the displays which result from a transform performed on a square wave of 512 points and from an inverse transform performed on the resulting coefficients.

4.1 Input Display

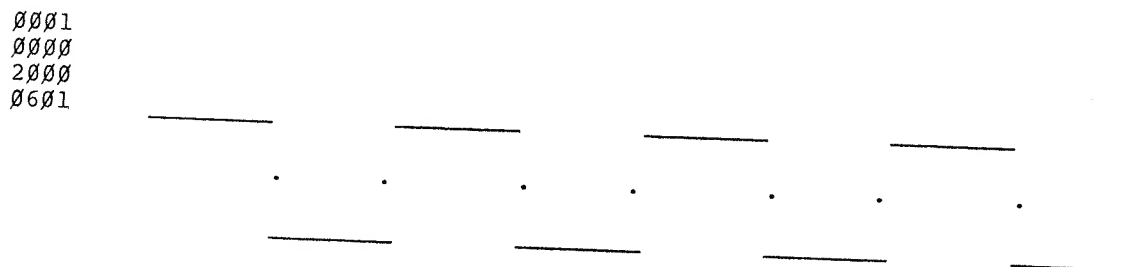
Consider a square wave¹ of 512 real points which has the following format on tape or disk:

Address	Value	
0	2000	77 points
77	1000	
100	0000	77 points
177	1000	

¹The displays shown on the following pages are adaptations and are for demonstration purposes only.

Address	Value	
277	2000	> 77 points
277	1000	
300	0000	> 77 points
377	1000	
400	2000	> 77 points
477	1000	
500	0000	> 77 points
577	1000	
600	2000	> 77 points
677	1000	
700	0000	> 77 points
777	1000	

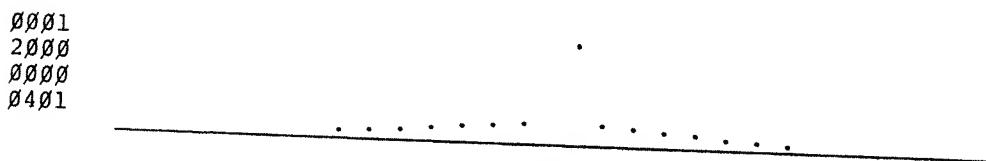
If the input is displayed, there will only be a REAL display. It will look as follows, assuming the cursor is to the extreme left and the display is not moving.



The first two values in the upper left hand corner are the address of the point on which the cursor is resting. When the cursor is at the extreme left, it indicates location 0000 of field 1. The third value is the contents of that memory location, in this case, 2000. The fourth value is the position of the cursor with respect to the bottom of the screen. [1 = bottom, 401 = X axis (middle), 1000 = top.]

4.2 Transform Displays

4.2.1 Real Display



Moving the cursor to the highest point in the display will change the value display to:

```

0001
2400
2000
0601

```

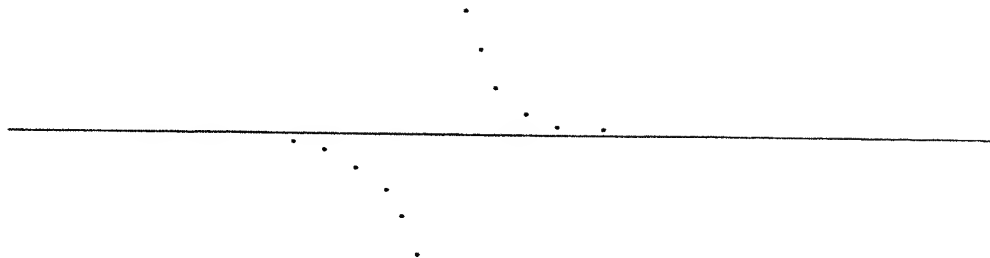
This is the DC component of the wave.

4.2.2 Imaginary Display

```

0001
2000
0000
0401

```



Moving the cursor to the lowest point produces the values:

```

0001
2374
6567
0257

```

Moving the cursor to the highest point displays:

```

0001
2404
1214
0522

```

4.2.3 Magnitude Display

```

0001
2000
0000
0401

```



Moving the cursor to the highest point gives the following display:

```

0001
2400
1000
0501

```

Because the magnitude of maximum values causes overflow, a factor of 2 is removed during computation. Therefore, the values displayed are half scale; type the key "1" once to display the magnitude at full scale.

4.2.4 Scale Factor Display

The scale factor has a value of 1. To obtain the actual coefficients, rest the cursor on the desired point and shift right the third value of the corner display the number of bits equal to the scale factor. In this example, the highest value of the real display is 2000. Shifting it right by the scale factor (=1) yields 1000, the actual value of the DC component, which in binary is 001 000 000 000. Because the binary point is to the right of the sign bit, the actual value is $+0.01_2$.

4.3 Inverse Transform Displays

The output of the transfer was 512 complex points. The inverse yields the following displays:

4.3.1 Real Display

```
0001
0000
0764
0477
```

.

The third value, 0764, is a deviation from 1000, the exact value. At this time there are 2 scale factors involved. The relationship between the computed results and the original data is:

$$\text{results} = [(\text{original data}) * 2^{\text{sum of scale factors}}] / \# \text{ of points}$$

Reducing the equation for the first point yields:

$$\begin{aligned} 1000_8 &= [(2000_8) * 2^8] / 1000_8 \\ 2^9 &= 2^{10} * 2^8 / 2^9 \\ &= 2^9 \end{aligned}$$

0001
1000
0007
0401

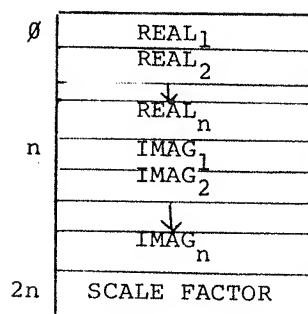
4.3.3 Magnitude Display

$\emptyset\emptyset\emptyset 1$
 $\emptyset\emptyset\emptyset\emptyset$
 $\emptyset 3 7 2$
 $\emptyset 4 4 \emptyset$

4.3.4 Scale Factor Display

5.0 DATA STORAGE

9



-only present if file is generated by the FFTD program.

6.0 DATA SCALING

All calculations in FFTD are done with single precision fixed point signed binary fractions. The binary point is located between bit 0 and bit 1, leaving an 11 bit signed mantissa. Bit 0 is used as a sign bit. Negative numbers are formed by taking the two's complement of the positive binary fraction, so all inputs must be scaled in magnitude to less than one. The outputs are also formatted as above.

In order to preserve precision, it is sometimes necessary to divide by 2 in a computation. As a result, a pseudo floating point format has been adopted in which a variable scale factor (or exponent) is imposed on all the Fourier coefficients. This scale factor or pseudo exponent is found in item SCAL after each transform has been completed. It is also stored after the last imaginary part on tape or disk. The values stored on tape or disk are the Fourier coefficients multiplied by 2^{SCAL} . Because in binary notation shifting a number right one bit is equivalent to dividing by two, to retrieve the coefficients themselves, shift each number right by the number of bits equal to the value of the scale factor. In the case of the inverse transform, the time samples are the values in memory multiplied by $2^{-\text{SCAL}}$. If, however, the inverse transform was performed on normalized transform data, the results are equal to $[(\text{original data}) * 2^n] / \text{no. of points}$ where n equals the sum of both scale factors. To retrieve the time samples, shift left each number by the value of the scale factor.

7.0 SUBROUTINES USED

Manipulation of the DIAL and data LINCtapes and disk is done using the program MILDRED (DEC-12-FZDA). The question and answer displays are handled by QANDA (DEC-12-FISA). The data displays are handled by DISPLAY

(DEC-12-FLSA). A modification of FFTS-C (DECUS #8-144) is used to perform the Fourier Transforms.

8.0 ALGORITHM DESCRIPTION

The Fast Fourier Transformation enables computation of the power spectrum of a time series in a minimum of time. Specifically, it permits the discrete Fourier transformation

$$S_j = \frac{1}{N} \left[\sum_{k=0}^{N-1} x_k e^{-2\pi i j k / N} \right] \quad \begin{matrix} j=0, \dots, N-1 \\ i = \sqrt{-1} \end{matrix}$$

of a series on N equally spaced time samples (where N is a power of 2). The time required is proportional to $N \log_2 N$, whereas previous methods required times proportional to N^2 . This gives a reduction in computation time of $1 - \log_2 N / N$ or over 99 percent for $N=1024$. The algorithm makes use of the fact that

$$W^k = W^{(k \bmod N)} \quad (\text{where } W = e^{-2\pi i / N})$$

to reduce the number of manipulations necessary for a transformation.

9.0 CORE CHART

Field 0

```

SEGMENT 0
    PAGE 0 - IFFT
    *4000 - FFT
    *14000 - DISPLAY
SEGMENT 1 - MILDRED
SEGMENT 2 - MONITOR
           QANDA
SEGMENT 3 - Data display code
           FDV table
           RWPARM table
           Questions
           Sine Table

```

Field 1

```

0 - Buffer - real parts
2000 - Buffer - imaginary parts

```

10.0 PROGRAM REGION DESCRIPTION

10.1 Routines

- 0117 IFFT - Take the Inverse Fourier Transformation of the data in field 1. The results are in bit inverted order (refer to the SORTX routine).
- 0400 FFT - Take the Fourier Transformation of the data in field 1. The results are in bit inverted order (refer to the SORTX routine).
- 0701 SORTX - Sort the data from bit inverted order to sequential order. Bit inversion means simply the process of re-ordering the bits in a binary number. For instance, the binary number 001 bit inverted is just 100 (=4). For example, to locate S_5 in memory for a 16 point transformation ($N=16, n=4$), write 5 as a binary number of $n=4$ bits, $5_{10}=0101_2$. Then reverse the order of these bits to 1010₂. This means S_5 is stored in position 10. Physically, then, S_5 of the real parts is to be found in location XRTAB+9.
- 1000 MULTIP - Perform a rounded single precision signed multiply using EAE. The CAL+1 contains the address of the multiplicand. The AC contains the multiplier. Exit with the product in the AC.
- 1010 INVRT - Reverse the bits of the number contained in the AC.
- 1060 TRIGET - Fetch sine and cosine values. Specifically, if the AC=K on entry, the values of $\sin(2\pi K/N)$ and $\cos(2\pi K/N)$ are fetched from an internal trig table. K must be $\geq N/2$. A register COSINE contains the cosine value and the AC contains the sine value on exit.
- ADDR - Perform a single precision add with rounding.

1200 IDORA - This subroutine generates a moving window display with a cursor riding on the curve. For more information refer to the DISPLAY document, DEC-12-FLSA-D.

4026 IFDIAL - Display the question: FROM DIAL UNIT? Y/N_ If the answer is Y, jump to UNTFIL; if N, jump to DATTAP; if neither, redisplay the question.

4044 UNTFIL - Jump to the subroutine ASK2 to display:

UNIT NUMBER__
FILE NAME_____

If the unit number is illegal, jump to ASK2 again to redisplay the question. If legal, jump to LOOKUP with the address of the File Description Vector (hereafter referred to as FDV) parameter list in the AC. If the file cannot be found, display the message:

CANNOT FIND
HIT RETURN TO CONT

When RETURN is hit, jump back to UNTFIL. If the file is found, jump to MOVINP.

4061 DATTAP - Jump to the subroutine ASK3 to display:

UNIT NUMBER__
BLK NUMBER___

If an illegal value is entered, jump back to DATTAP. If all the input is legal, fall through to MOVINP.

4063 MOVINP - Jump to FDV2RW to move the input information from the FDV to the read/write parameter list. Fall through to PTS.

4064 PTS - Display: NUMBER OF PTS
(4-1024 BY POWERS OF 2)
REAL OR
COMPLEX? R/C_

Set B1 to the address of the answer buffer, MPLIER to 12 and UPLEGL to -71 (-9) because the number of points is entered as a decimal value. Set the AC to the largest legal value, 20000, and jump to CONV. If the answer is an illegal value jump back to PTS; store the value in N and store its 1's complement in TEMP1. Since the number of points must be an integral power of 2, only one bit in TEMP1 may be set. Bit 11 is the exception to one bit being a power of 2. Check bit 11 first, then rotate the value adding up the number of bits set. If the total is not 1, jump back to PTS. Otherwise fall through to ROT1.

- 4136 ROT1 - Compute the power of 2 by rotating right the value in TEMP1 and stepping B2 until the bit that is set is encountered in bit 11. Fall through to STAMU.

- 4144 STAMU - Store the power of 2 in NU. If the power is less than 2, jump back to PTS. Otherwise load the AC with the number of points*2 and jump to NUMBKS to compute the number of blocks needed to hold the output. Store the value in FDV+7. Store it also in RWPARM+3 since, for complex data, the input and output data consist of the same number of blocks. If the answer to the second question is not R, jump to IFCOM. If it is R, the input consists of half as many words as the output. Load the AC with the value of N and jump to NUMBKS to compute the number of input blocks. Store the value in RWPARM+3. Set REALFG and jump to CKEND.

- 4023 IFCOM - If the answer is C, clear REALFG and fall through to CKEND. Otherwise jump back to PTS to redisplay the question.

- 4211 CKEND - If there is not enough room between the starting block number and the end of tape to hold the number of points specified, jump back to PTS. If

the number of output words is 4000 or greater,
another block will be needed to hold the scale
factor. Increment FDV+7. Fall through to IFFFT.

4231 IFFFT - Display: FFT OR DISPLAY? F/D_
TRANSFORM OR
INVERSE? T/I_

If the answer to the first question is D, set
DISFLG to indicate that the data will only be
displayed. If F, clear DISFLG to indicate that
a Transform or Inverse Transform will be performed.
If the answer to the second question is T, clear
FTFLG; if I, set it. If DISFLG is set, jump to
DISPLY to display the data. Otherwise, jump to
OUTQES.

4273 OUTQES - Display the question: OUTPUT ON DIAL UNIT? Y/N_
If the answer is Y jump to OUTUNT; if N jump to
ONDAT; otherwise redisplay the question.

4310 OUTUNT - Jump to the subroutine ASK2 to display:

UNIT NUMBER__
FILE NAME_____

If an illegal value is input, redisplay the ques-
tion. Otherwise jump to ENTER with the address of
the parameter list in the AC. If a file with the
specified name already exists, jump to SAMNAM. If
there is not enough space to hold the output data,
jump to NOSPACE. If it is a new file and there is
enough space to hold it, fall through to RDDATA.

4320 RDDATA - Clear 4000 words of field 1 and read in the input
data. If REALFG is 0, the data is complex - move
the imaginary parts to start at location 2000. If
it is non-zero, the data is real and nothing need
be done. Jump to PROC.

4357 PROC - If IFTFLG is 0, jump to FT to do a Transform.
Otherwise, fall through to do an Inverse Transform.

- 4363 IFT - Jump to the subroutine IFFT to do an Inverse Transform on the input data. Then jump to the subroutine SORTX to sort the coefficients into sequential order from bit inverted order. Jump to STSCAL to store the scale factor which is equal to NU-SCAL. The data should be shifted by this value.
- 4365 FT - Jump to the subroutine FFT to transform the input data. Then jump to the subroutine SORTX to sort the coefficients into sequential order from bit inverted order. The scale factor is the value in SCAL and equals the number of bits by which the data should be shifted right. Fall through to STSCAL.
- 4367 STSCAL - Store the scale factor in the word following the last imaginary part. Move the imaginary parts from 2000 to immediately behind the real parts.
- 4377 NOWSTR - Jump to the subroutine FDV2RW to move the output parameters from the FDV to the read/write parameter list. Write the data onto the output tape and jump to DISPLY.
- 4423 NOSPAC - Jump to the subroutine ASK to display the message:

NO SPACE
HIT RETURN TO CONT

When RETURN is hit, jump to OUTQES.

- 4430 SAMNAM - Jump to the subroutine ASK to display:

ALREADY EXISTS
REPLACE? Y/N_

If the answer is Y, jump to REPL; if it is N, jump to OUTUNT. If it is neither, redisplay the question.

4446 REPL - Try to replace the existing file with the new file. If the new file is longer, jump to NOSPAC. If the replacement is successful, jump to RDDATA.

4452 ONDAT - Jump to the subroutine ASK3 to display:

UNIT NUMBER__
BLK NUMBER___

If an illegal value is entered, redisplay the question. If there is not enough space between the specified block number and the end of tape to hold the output data, jump to NOSPAC. Otherwise, jump to RDDATA.

10.2 Subroutines

4466 FDV2RW - Transfer the unit number, starting block number, and number of blocks from the FDV parameter list to the READ/WRITE parameter list.

4503 NUMBKS - Enter with the number of words in the AC. Convert this value to blocks by counting the number of times 400 can be subtracted from it before the value becomes negative. Return with the number of blocks in the AC.

4523 ASK2 - Jump to OCTL to set MPLIER to 10 and UPLEGL to -67(-7) because the unit number is input as an octal number.

Display: UNIT NUMBER__
FILE NAME_____

by jumping to the subroutine ASK with the address of QUES2 in the AC. Set B1 to the address of the answer buffer and jump to the subroutine CONV with the largest legal unit number, 17, in the AC. If the value is illegal, return to CALL+1. If legal, store it and the file name in the FDV parameter list. Fill the file name out to 8 characters with 77's. Return to CALL+2.

4572 ASK3 - Display: UNIT NUMBER __
BLK NUMBER _____

by jumping to the subroutine ASK with the address of QUES3 in the AC. Set B1 to the address of the answer buffer and jump to OCTL to set MPLIER to 10 and UPLEGL to -67(7) because the unit and block numbers are input in octal. Jump to subroutine CONV with the largest legal unit number, 17, in the AC. If the value is illegal, return to CALL+1. Otherwise, store it in word 0 of the FDV parameter list. B1 is now pointing to the block number. Jump to CONV with the largest legal block number, 777, in the AC. If the value is illegal, return to CALL+1. If legal, store it in word 6 of the FDV parameter list. Return to CALL+2.

4627 CONV - CONV is entered with the largest legal value in the AC and B1 pointing to the address - (1 half word) of the first character to be converted. Store the 1's complement of the largest legal value in TEMP2 and clear TEMP1. UPLEGL contains a -71(-9) or -67(-7) and MPLIER contains a 10 or 12 depending on whether the number to be converted is in decimal or octal. Extract a character and compare it against an ASCII 0 and the contents of UPLEGL. If it is a legal value, jump to MULPLY which will multiply the value in TEMP1 by the contents of MPLIER and add the digit being converted to it. Repeat the procedure until a character is found which is not between 0 and UPLEGL. If it is not a 34, 74, or 0, it is an illegal character: return to CALL+1. A 34 or 74 indicates the end of the input field; a 0 indicates the end of the input. Compare the converted value in TEMP1 against the maximum legal value in TEMP2. If the value is legal return to CALL+2; otherwise return to CALL+1.

4711 OCTL - OCTL sets MPLIER to 10 and UPLEGL to -67(-7) so that CONV will convert an octal number.

- 4720 ASK - ASK is entered with the address of the display in the AC. Store it in the parameter list and jump to QAINIT to display the message. Refresh the display until the answer is input. Return to the calling routine.
- 6001 DISPLY - This region is entered either after the Transform or Inverse Transform is completed or in response to a D in answer to the display: FFT OR DISPLAY? F/D_. Since the data is manipulated in preparation for each display it must be read in before each display. After reading in the data, display:
- WHICH DISPLAY?
R(EAL)
I(MAGINARY)
M(MAGNITUDE)
S(SCALE FACTOR)
LINE FEED (RESTART)
- If the answer buffer contained Ø, just LINE FEED was hit: jump to IFDIAL to restart the program. Otherwise jump to WCHDIS.
- 6035 WCHDIS - Jump to DPIMAG, DPMAG, DPREAL, or DPSCAL if the answer was I, M, R, or S, respectively. Otherwise redisplay the question.
- 6055 DPIMAG - If REALFG is non-zero, the input is real and no Transform was performed. Therefore, there are no imaginary parts to display; redisplay the question. If REALFG is zero, check IFTFLG. If it equals zero, either an Inverse Transform was performed or the original data is just being displayed. In either case the data is in the right order. If IFTFLG is non-zero, a transform was performed. The positive half of the curve is first followed by the negative half and the signs are reversed. Swap the halves and reverse signs before jumping to PREPAR.
- 6117 DPREAL - Check IFTFLG for the same reason as in DPIMAG. The only difference is that the signs of the real parts are not reversed.

- 6130 PREPAR - If less than 1000 points are to be displayed, the display will not move and the points displayed will be centered on the scope. To achieve this, LEFTX is set to the 1's complement of $-1000 + (1000 - \# \text{ of points}) / 2$, MINPTS to the 2's complement of the number of points, and MVDIS to the instruction CLR. Jump to SHOWIT.
- 6147 GQ1000 - If 1000 or more points are to be displayed, the display will fill the scope and will move. To achieve this, LEFTX is set to the 1's complement of 1000, MINPTS to the 2's complement of 1000 and MVDIS to the instruction SCR 4. Fall through to SHOWIT.
- 6162 SHOWIT - Jump to the subroutine IDORA to display the data. The six parameters following the call to IDORA are in order: the memory field of the lower address, the lower address, the memory field of the higher address, the higher address, the Y offset of the display and the scale factor of the data. Both fields are always 1, the lower address is always 0. The higher address is set in the region DISPLY. The Y offset is always 0; therefore the baseline is half way up the scope. The scale factor is the instruction SCR plus the number of bits to scale the data right before displaying it. Since IDORA displays only the right nine bits, if the left three bits are significant, the data must be scaled right three before displaying it.
- 6171 RFRSH - Jump to RDORA to refresh repeatedly the display until a key on the teletype is hit. If the RETURN is hit, jump to REDPLY which jumps to DISPLY to redisplay the question: WHICH DISPLAY? If a 1 is entered, jump to LARGER to blow up the display. If a Q is hit, jump to SMALLR to decrease its size. If anything else is entered, ignore it.
- 6211 SMALLR - If the instruction at SIZE contains a shift of 11 bits, a bigger shift would be meaningless. Jump back to RFRSH. Otherwise, increment the value of the shift and jump to SHOWIT.

- 6216 LARGER - If the instruction at SIZE contains a shift of 0 bits, jump back to RFRSH. Otherwise decrement the value of the shift and jump to SHOWIT.
- 6226 DPSCAL - If REALFG is non-zero, only real parts are present, meaning this program did not create the file and therefore there is no scale factor. Return to DISPLY to redisplay the question. If REALFG is 0, the scale factor is stored after the last imaginary part. Convert it to ASCII decimal and display it.
- 6270 DPMAG - If REALFG is non-zero, the input data is real and no transform was performed; therefore the magnitude is the same as the real points. Redisplay the question: WHICH DISPLAY? Otherwise move the imaginary parts to location 2000. Set RELPTR and IMGPTR, which contain the effective address of the multipliers, to 6000 since the data begins at location 0 of their respective segments and is fractional. Fall through to NXTMAG.
- 6320 NXTMAG - Square a real part and store it. Square the imaginary part, add the square of the real part to it, jump to the subroutine SQRT to get the square root of the sum and store it in place of the real part. Repeat the process for each point. Then jump to SHOWIT to display the magnitude.
- 7116 MOVPTS - The subroutine MOVPTS moves values from one buffer (address -1 in l0) in field 1 to another (address -1 in l1). If CMPFLG equals 1, the values are complemented as they are moved. TEMPR contains the 2's complement of the number of values to move.
- 7132 MVRLMG - The subroutine MVRLMG is used to swap the first and second halves of the real or magnitude values. In the process they are moved from the buffer starting at location 0 to the one starting at 2000.
- 6375 FDV - The File Descriptor Vector parameter list is used by the LOOKUP, ENTER, and REPLACE sections of MILDRED. Word 0 contains the unit number, words 1-4 contain

the file name, word 5 contains a 2 indicating the file is binary, word 6 is the starting block number, and word 7 is the number of blocks. Word 6 is filled by LOOKUP, ENTER and REPLACE. Word 7 is filled by LOOKUP but must be supplied for ENTER and REPLACE.

6405 RWPARM - The Read/Write parameter list is used by the READ and WRITE sections of MILDRED. Bits 0-2 of word 0 contain the field, bits 9-11 contain the unit. Word 0 contains the starting address, word 1 the starting tape block number and word 2 the number of blocks.

7052 SQRT - The subroutine SQRT is entered with a value in the double precision location DPSQ. It returns with the square root in the AC.

10.3 Symbols

N	Number of words in computation
NU	Power of 2 of value of N
L	Index to show what array is being constructed
S	Gives spacing between node pairs in the Lth array
NOVER4	Storage for N/4
MAXNU	Power of 2 of largest table size (13)
MNOVR2	Storage for N/2
QR	Pointer to real part of X(Q)
QI	Pointer to imaginary part of X(Q)
PR	Pointer to real part of X(P)
PI	Pointer to imaginary part of X(P)
Q	Numerical index Q ($=0, 1, \dots, N-1$)
P	Numerical index P ($=0, \dots, N-1$)
K	Number in the node being operated on
C	Interrupts computation of Lth array every S passes
ADD2	Used by subroutine ADDR as data (addend) Used by monitor as a temporary location
TEMPR	Temporary storage register for real parts Used by monitor as a temporary location
SINE	Temporary storage for $\sin(S*PI*K/N)$ Used by monitor as a temporary location
COSINE	Temporary storage for $\cos(2*PI*K/N)$ Used by monitor as a temporary location
GR	Real part of product ($W^k * X(P)$) - temporary storage Used by monitor as a temporary location
GI	Imaginary part of product ($W^k * X(P)$) - temporary storage
SCAL	Pseudo exponent of Fourier coefficients
SHFLAG	If $=1$, add with shift; if $=0$, add without shift
SHFCHK	Indicates if all X's in an iteration are $<.5$
DISFLG	If $\neq 0$, the data will just be displayed
IFTFLG	If $\neq 0$, an Inverse Transform was performed
REALFG	If $\neq 0$, the data does not contain imaginary parts
DPSQ	Used to save the double precision squares of the real and imaginary parts during calculation of the magnitude.
CMPFLG	If $=1$, the subroutine MOVPTS will complement the values as it moves them

10.4 Beta Registers

Beta registers 1, 2, and 3 are used by the monitor in ASK2 and ASK3 as temporary pointers and counters.. QANDA and MILDRED make more extensive use of the Beta registers.

11.0 ASSEMBLY INSTRUCTIONS

The FFTD program is assembled in three sections by assembling and saving each, then adding them together. The entire command sequence is:

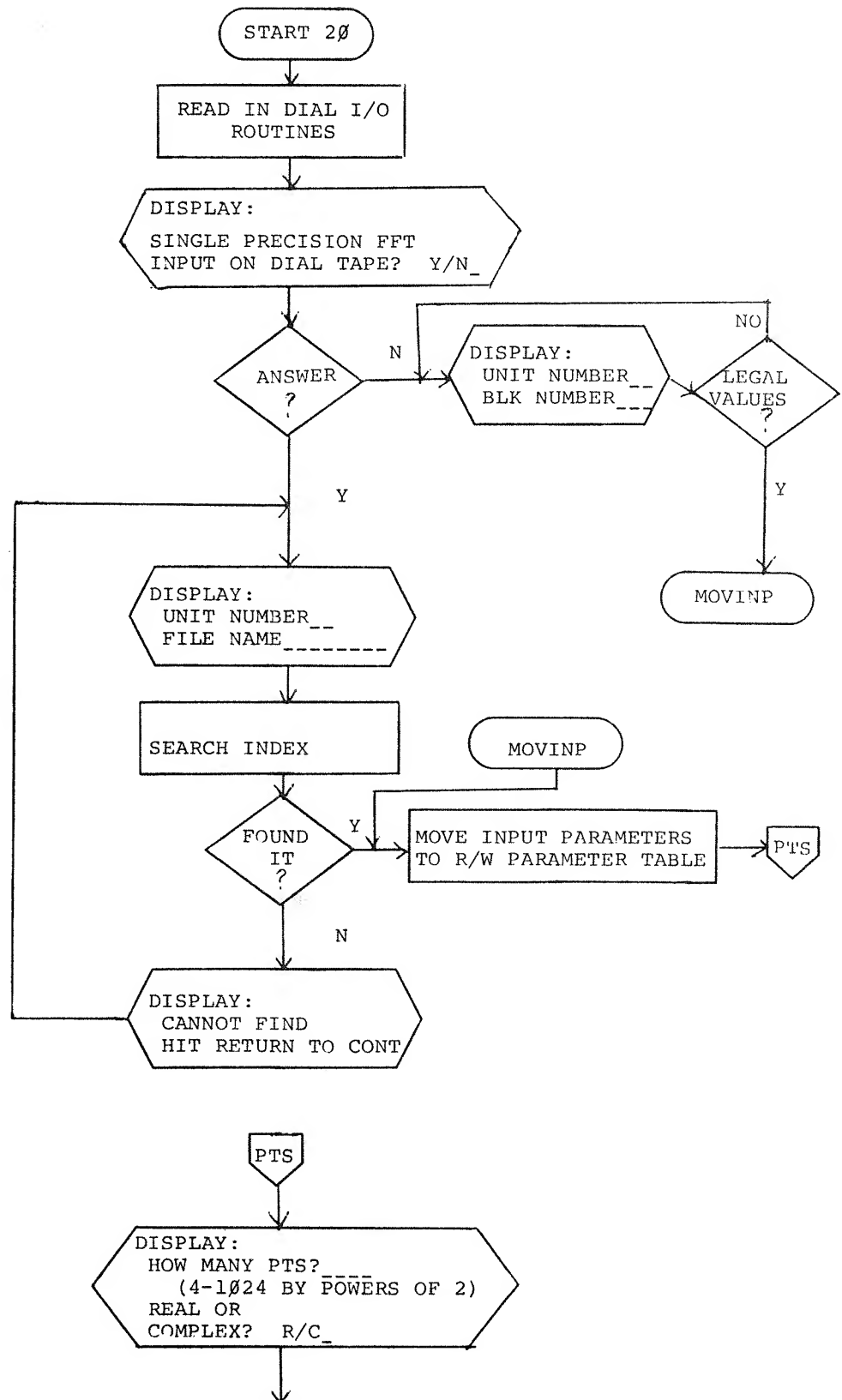
→AS MILQAN,n }	
→SB MILQAN,n }	
→AS SIN256,n }	
→SB SIN256,n }	
→AS FFTC-1 }	
→SB FFTC-1 }	(FFTC-1 chains to FFTC-2)
→ZE }	
→AB MILQAN,n }	
→AB SIN256,n }	
→AB FFTC-1,n }	
→SB FFTD,n,L }	(saves the whole program)

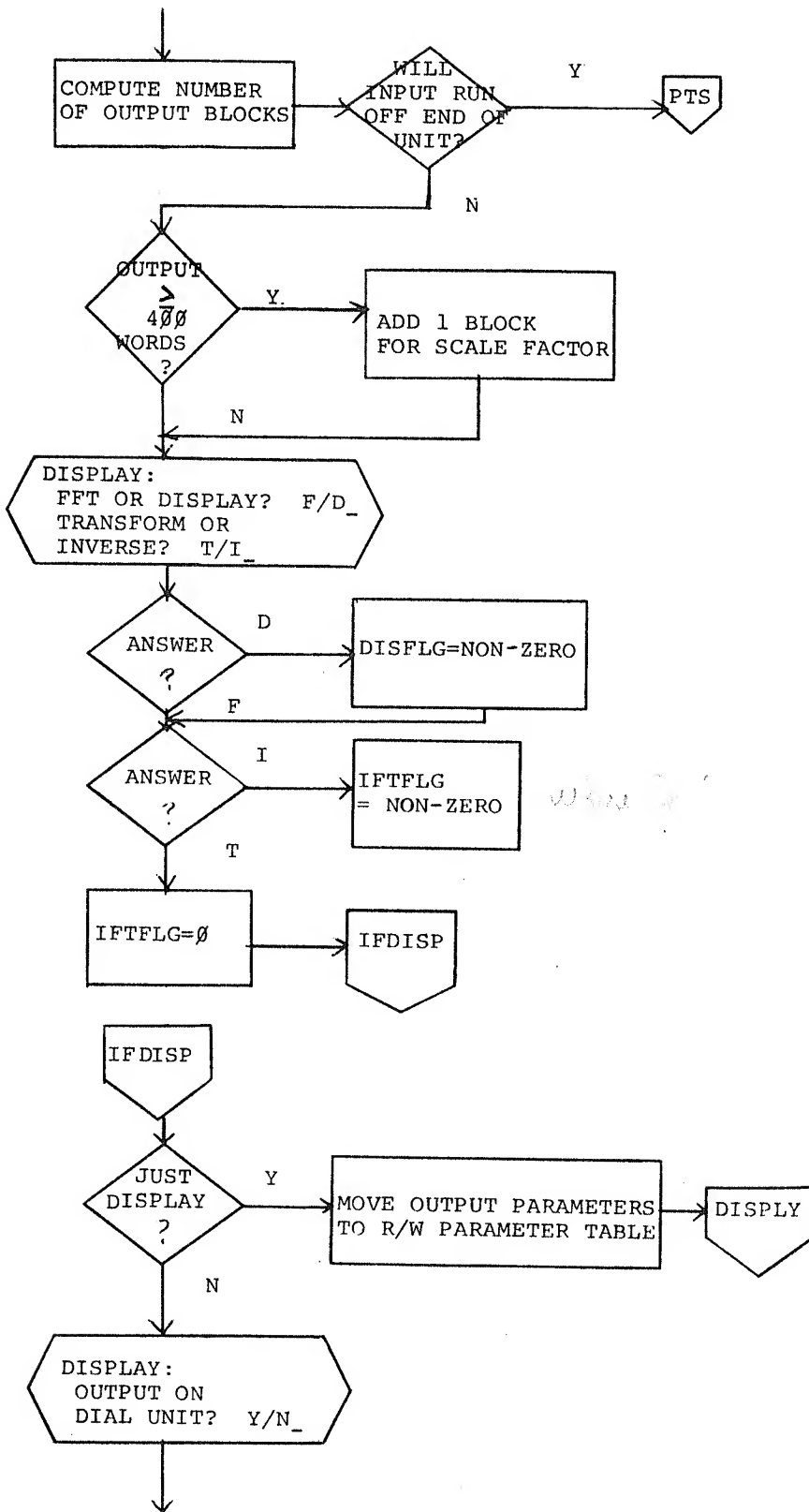
12.0 SYSTEM FLOWCHARTS

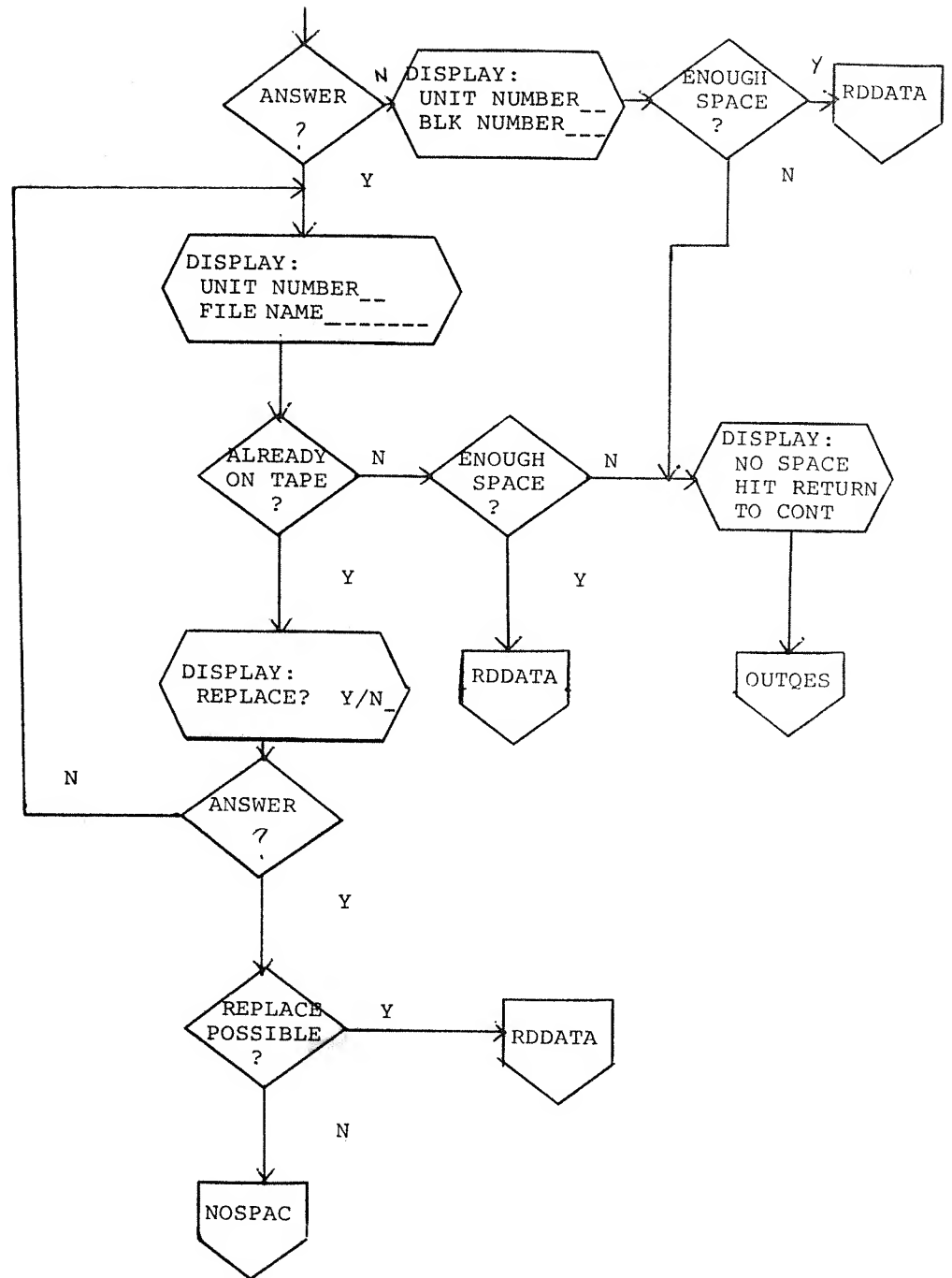
(Attached)

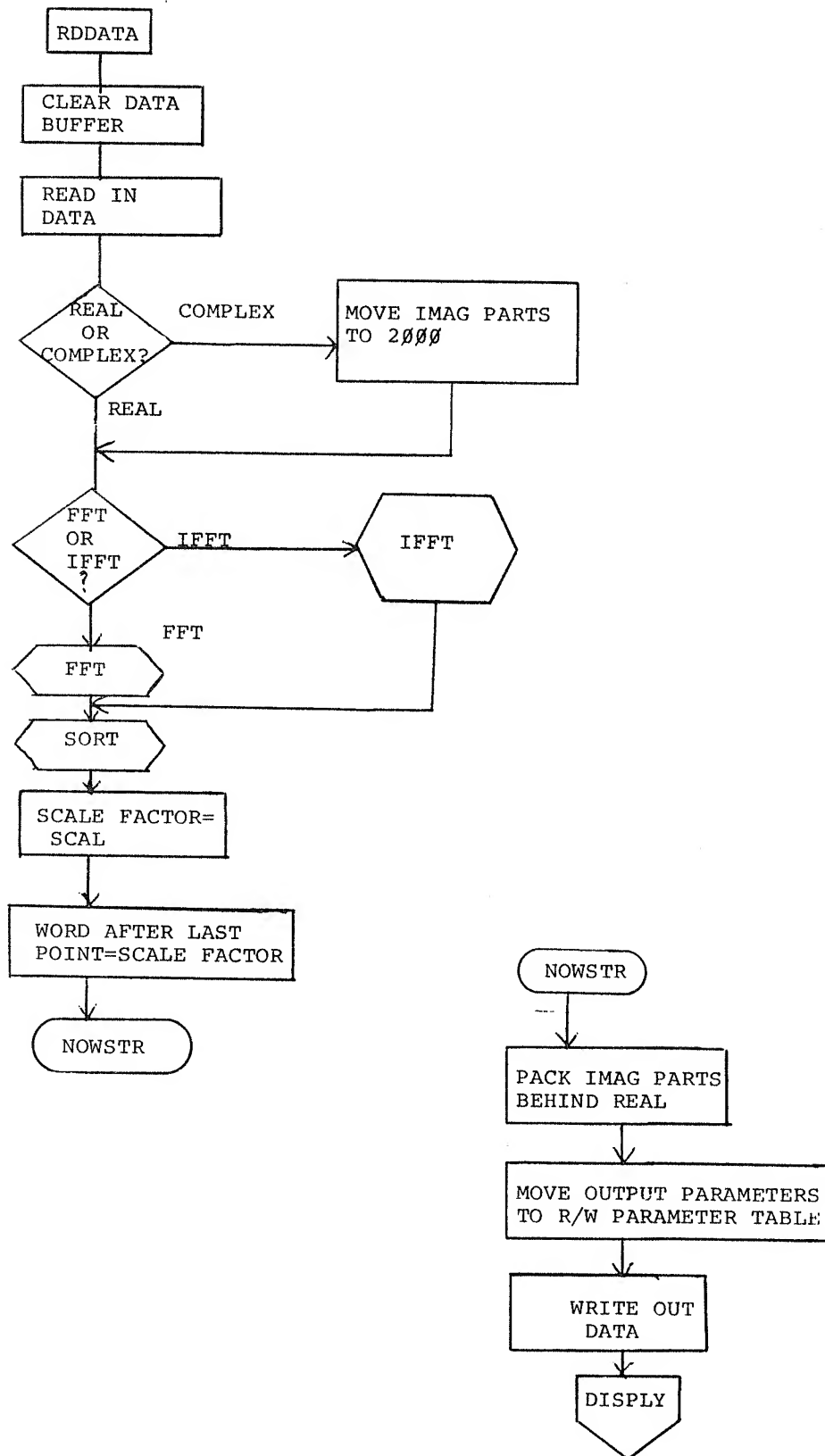
13.0 PROGRAM LISTING

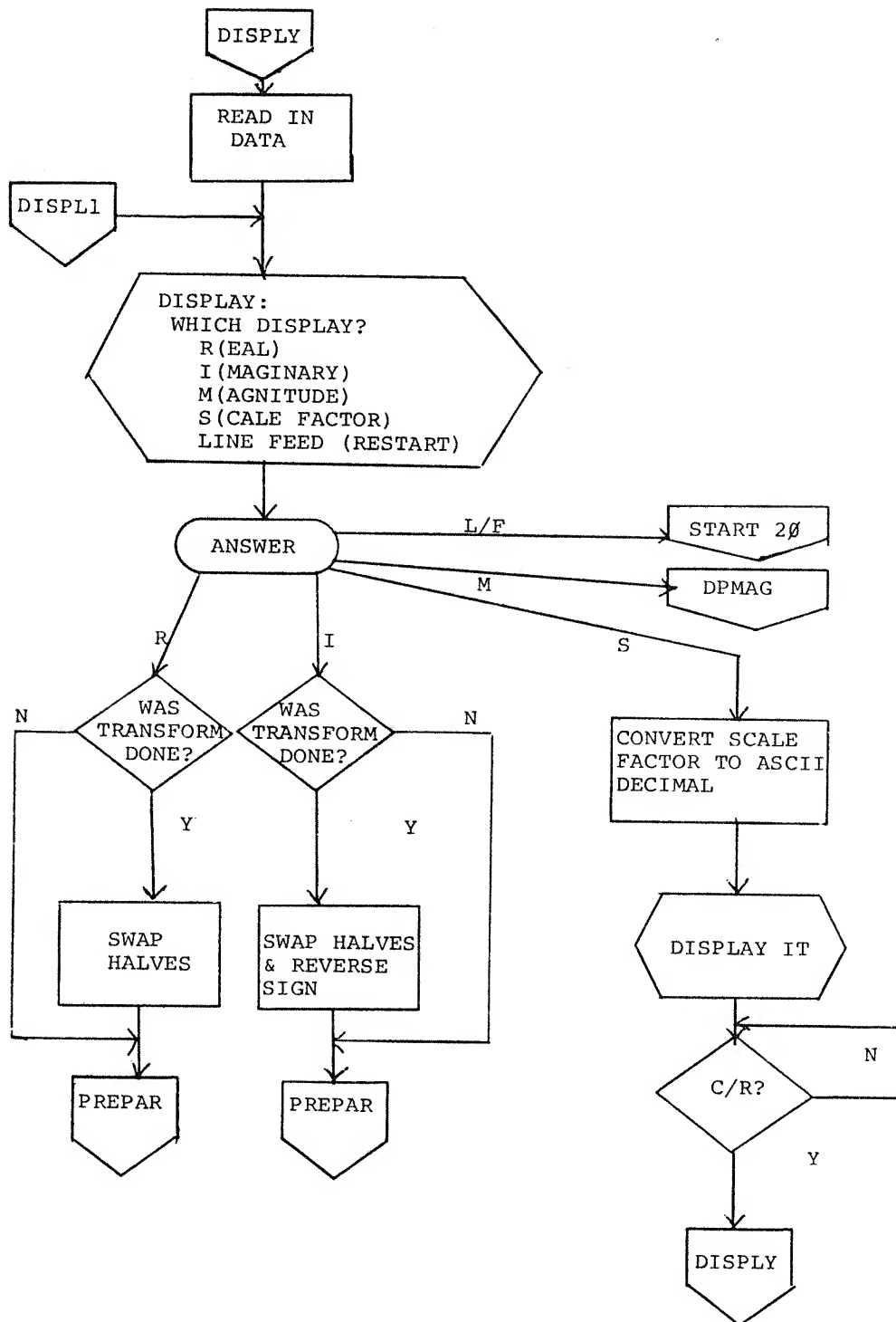
(Attached)

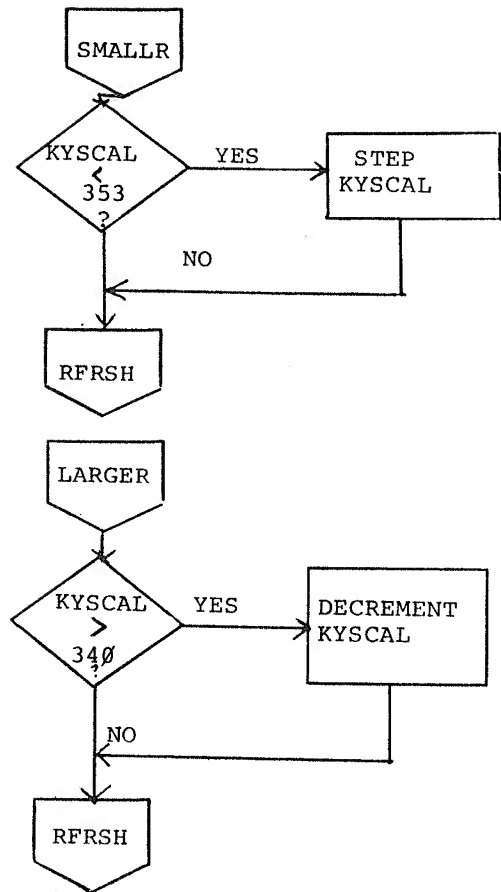
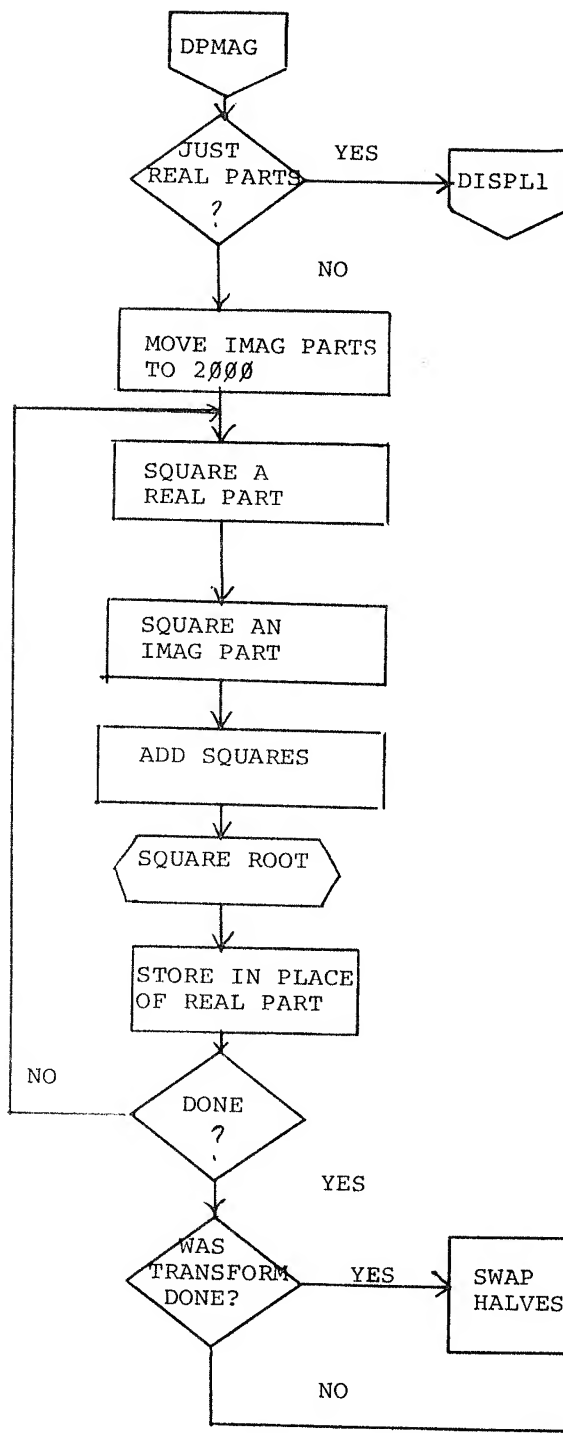


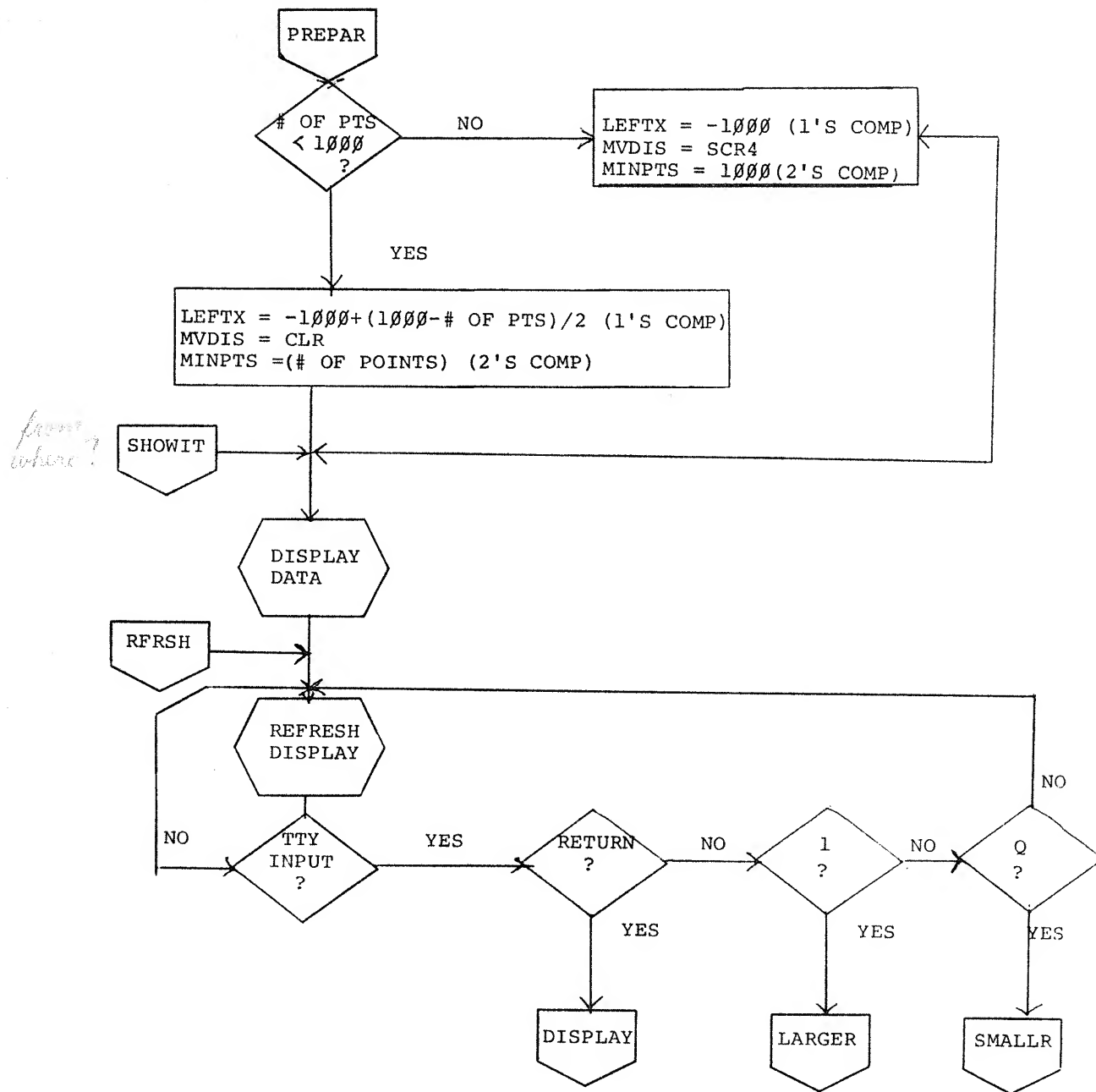













```

00000
00001
00002
00003
00004
00005
00006
00007
00008
00009
00010
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00013
00014
00015
00016
00017
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00064
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00070
00071
00072
00073
00074

*20
/FFTS=REAL
/THIS IS A PROGRAM FOR CALCULATING THE
/FAST FOURIER TRANSFORMATION OF N REAL
/TIME SAMPLES WHICH ARE STORED ON DISK
/OR DATA TAPE OR DISK
/TO BE RUN ON A PDP-12 COMPUTER EQUIPPED WITH THE FOLLOWING MINIMUM HARDWARE:
/ 1) ASR 33 OR ASR 35 TELETYPE
/ 2) 8 K OF CORE MEMORY
/ 3) VR12 CRT DISPLAY
/
/COPYRIGHT 1970, DIGITAL EQUIPMENT CORPORATION
/ MAYNARD, MASS, 01754
/TRANSFORM ALGORITHM
/WRITTEN BY JAMES ROTHMAN -- AUGUST, 1968
QARFSH=1053
GAINIT=1000
XRTAB=0
XITAB=2000
SINTAB=7347
CDF1=6211
CDF0=6201
PMODE
/PAGE ZERO
*3
/TABLE PARAMETERS
N, 0000
NU, 0000
L, 0000
S, 0000
F, 0000
/NUMBER OF POINTS IN COMPUTATION DIVIDED BY 2
/POWER OF TWO OF POINTS IN COMPUTATION (N=2*NU) MINUS 1
/INDEX TO SHOW WHAT ARRAY IS BEING CONSTRUCTED
/GIVES SPACING BETWEEN NODE PAIRS IN THE LTH ARRAY.
/USED FOR SCALING NODE POSITION TO GET NUMBER IN NODES.
/STORAGE FOR N/4
/LARGEST TABLE SIZE (POWER OF 2)
/STORAGE FOR -N/2
/POINTER TO REAL PART OF X(Q)
/POINTER TO IMAG, PART OF X(Q)
/POINTER TO REAL PART OF X(P)
/POINTER TO IMAG, PART OF X(P)
/NUMERICAL INDEX Q(=0,1,...,N-1)
/NUMERICAL INDEX P(=0,1,...,N-1)
/NUMBER IN THE NODE BEING OPERATED ON
/INTERRUPTS COMPUTATION OF LTH ARRAY EVERY S PASSES
/USED BY SUBROUTINE ADDR AS DATA (ADDEND)
/TEMPORARY STORAGE REGISTER FOR REAL PARTS
/TEMP. STORAGE FOR SIN (S*PI*K/N)
/TEMP. STORAGE FOR COS (2*PI*K/N)
/REAL PART OF PRODUCT (W*K)*X(P), TEMP STORAGE
/IMAG. PART OF (W*K)*X(P), TEMP STORAGE
/ADD C(AC) TO C(ADD2) AND SCALE RIGHT ONE IF NECESSARY.
/BIT INVERTED BUFFER SORTED
/WORD IN AC OF NU BITS IS BIT INVERTED
/FETCH SIN AND COS OF 2*PI*C(AC)/P
/DO FFT OF THE INPUT BUFFER
/DO INVERSE OF BUFFER

```

```

0076 0077 0100 0101 0102 0103 0104 0105 0106 0107 0110 0111 0112 0113 0114
0051 0000 0000 0000 0001 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0052 0000 0000 0000 0001 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0053 0000 0000 0000 0001 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0054 0001 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0055 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0056 1077 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0057 1114 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0060 1125 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0061 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0062 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0063 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0064 0544 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0065 0542 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0066 0515 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0067 4000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0070 7777 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0071 7766 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0072 7770 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0073 6160 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0074 4060 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0075 7774 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0076 6270 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0077 7767 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0100 7773 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0101 6000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0102 7563 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0103 7457 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0104 7425 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0105 7440 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0106 7517 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0107 7400 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0110 1777 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0111 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0112 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0113 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0114 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0115 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0116 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0117 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0118 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0119 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0120 1343 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0121 6162 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0122 6171 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0123 6404 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0124 1361 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0125 1363 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0126 7132 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0127 7116 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0130 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0131 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0132 6053 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0133 4356 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0134 7774 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0135 7775 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0136 1444 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0137 1000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0140 2000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0141 6777 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0142 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0143 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0144 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0145 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0146 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0147 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0150 1200 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0151 1343 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0152 6162 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0153 6171 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0154 6404 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0155 1361 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0156 1363 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0157 7132 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0160 7116 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0161 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0162 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0163 6053 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0164 4356 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0165 7774 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0166 7775 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0167 1444 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0170 1000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0171 2000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0172 6777 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
0173 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000

```

```

/INPUT BUFFER AND /B
/DIFF IN ADDR OF REAL & IMAG PART TABLE
/PSEUDO FLOATING POINT FORMAT FLAGS
/PSEUDO EXPONENT OF FOURIER COEFFICIENTS
/IF =1, ADD WITH SHIFT; IF=0, ADD WITHOUT SHIFT
/INDICATES IF ALL XS IN AN ITERATION ARE <.5
/POINTERS TO SINE TABLE LOOK-UP SHIFTS
/THE NUMBER 10-NU MUST BE PLACED
/IN EACH OF THESE LOCATIONS
/POINTERS TO INSTRUCTION "FLAG" LOCATIONS

```

0174
0175
0176
0177
0200
0201
0202

0143 0000
0144 0644
0145 0344
0146 0011

LDF4,
SCR4,
CCLR,

0 LMODE
LDF 4
SCR 4
CLR
PMODE
EJECT

```

0203 /THIS SUBROUTINE TAKES THE INVERSE FFT (IFFT) OF THE DATA IN THE BUFFER.
0204 /IT IS ASSUMED THAT THIS DATA IS STORED SEQUENTIAL ORDER.
0205 /THE RESULTS ARE STORED IN BIT INVERTED ORDER.
0206 /THE ALGORITHM USED IS AS FOLLOWS:
0207 /   THE NORMAL TRANSFORM IS PERFORMED, EXCEPT:
0210 /   ON FETCHING THE VALUE FOR IM[W*K], WHICH IS
0211 /   THE SIN(2*PI*K/N), THIS SIN VALUE IS NEGATED.
0212 /
0213 /THE REASONING FOR THIS IS AS FOLLOWS:
0214 /   A WEIGHTING FACTOR OF W*(8-K) IS USED IN THE IFFT
0215 /   AND SINCE W*K AND W*(-K) ARE THE SAME EXCEPT THAT
0216 /   THEIR IMAGINARY PARTS HAVE OPPOSITE SIGNS, IT FOLLOWS
0217 /   THAT IM[W*K] SHOULD BE REPLACED BY -IM[W*K].
0220 /
0221 /
0222 /
0223 /
0224 /
0225 /
0226 /
0227 /
0230 /
0231 /
0232 /
0233 /
0234 /
0235 /

0147 0000 IFFT,
0150 7500 CLA CLL
0151 1152 TAD CCIA /NEGATE IM[W*K], GET CIA INSTRUCTION
0152 3561 DCA I SGNADJ /AND PUT AT LOCATION ADJSN
0153 4446 JMS I DOFFT /DU FFT
0154 6201 CDF0
0155 1163 TAD CNOP /RE-INSTATE NOP AT ADJSGN FOR FFT,
0156 3561 DCA I SGNADJ
0157 6211 CDF1
0160 5547 JMP I IFFT /EXIT
0161 0570 SGNADJ, ADJSGN /POINTER TO SIGN ADJUST INSTRUCTION
0162 7041 CCIA, CIA
0163 7000 CNOP, NOP
      EJECT

```

```

0236      *400      /COMPUTATION OF FIRST COMPLEX ARR FROM INPUT DATA
0237      /NUMBER OF INPUT POINTS IN "N" .L 2(N)IN"NU", FOR DETAILS OF ALGORITHM, SEE FLOWCHA
0238      FFT,
0239      0000
0240      0400
0241      0241 7301
0242      0401 7301
0243      0402 3005
0244      0403 3053
0245      0404 7001
0246      0405 3054
0247      0406 3055
0248      0407 1003
0249      0410 7112
0250      0411 3020
0251      0412 1004
0252      0413 7041
0253      0414 1021
0254      0415 3456
0255      0416 1456
0256      0417 3457
0257      0420 1457
0258      0421 3460
0259      0422 1003
0260      0423 7110
0261      0424 3006
0262      0425 1006
0263      0426 7041
0264      0427 3022
0265      0430 7040
0266      0431 1006
0267      0432 1051
0268      0433 3023
0269      0434 1004
0270      0435 7041
0271      0436 7001
0272      0437 3007
0273      0440 1023
0274      0441 1006
0275      0442 3025
0276      0443 1023
0277      0444 1052
0278      0445 3024
0279      0446 1025
0280      0447 1052
0281      0450 3026
0282      0451 6211
0283      0452 1424
0284      0453 3033
0285      0454 1426
0286      0455 4441
0287      0456 3034
0288      0457 1424
0289      0460 3033
0290      0461 1426
0291      0462 7041
0292      0463 4441
0293      0464 3426
0294      0465 1034
0295      0466 3424
0296      0467 1423
0297      0470 3033
0298      0471 1425
0299      0472 4441
0300      0473 4441
0301      0474 4441
0302      0475 4441
0303      0476 4441
0304      0477 4441
0305      0478 4441
0306      0479 4441
0307      0480 4441
0308      0481 4441
0309      0482 4441
0310      0483 4441
0311      0484 4441
0312      0485 4441
0313      0486 4441
0314      0487 4441
0315      0488 4441
0316      0489 4441
0317      0490 4441
0318      0491 4441
0319      0492 4441
0320      0493 4441
0321      0494 4441
0322      0495 4441
0323      0496 4441
0324      0497 4441
0325      0498 4441
0326      0499 4441
0327      0500 4441
0328      0501 4441
0329      0502 4441
0330      0503 4441
0331      0504 4441
0332      0505 4441
0333      0506 4441
0334      0507 4441
0335      0508 4441
0336      0509 4441
0337      0510 4441
0338      0511 4441
0339      0512 4441
0340      0513 4441
0341      0514 4441
0342      0515 4441
0343      0516 4441
0344      0517 4441
0345      0518 4441
0346      0519 4441
0347      0520 4441
0348      0521 4441
0349      0522 4441
0350      0523 4441
0351      0524 4441
0352      0525 4441
0353      0526 4441
0354      0527 4441
0355      0528 4441
0356      0529 4441
0357      0530 4441
0358      0531 4441
0359      0532 4441
0360      0533 4441
0361      0534 4441
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0336 /GET RELX(Q)]
0337 /RE=REAL PART
0340 /FORM RELX(Q)-(P)] (DIVIDED BY 2)
0341 /PUT AT RELX(P)]
0342 /GET RELX(Q)+X(P)]
0343 /PUT AT RELX(Q)],REAL PARTS DONE
0344 /Q=QR-XRLOC
0345
0346 /AC IS Q
0347 /IS Q>0? (IE THE WHOLE ARRAY HAS NOT BEEN COVERED)
0350 /NO, Q=0, DONE WITH FIRST ARRAY, MOVE ON TO OTHERS
0351 /YES, Q<0-1, MOVE UP THIS ARRAY
0352 /OR EQUIVALENTLY, QR<QR-1
0353
0354 /DO NEXT NODE PAIR
0355 /L GIVES THE NUMBER OF THE VERTICAL ARRAY JUST BUILT
0356
0357 /IS L=NU? (IE HAS THE LAST ARRAY BEEN COMPUTED?)
0360
0361 /YES, DONE, RESULTS STORED IN BIT REVERSED ORDER
0362 /GET SCALE FACTOR AND ADJUST FOR PROPER
0363 /ADDITION ON NEXT ITERATION
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TAD I WK
DCA ADD2
TAD I PR
CIA
JMS I ADDER
DCA I PR
TAD TEMPR
DCA I QR
TAD XRLOC
CIA
TAD QR
SPA SNA CLA
JMP CHKPT
CMA
TAD QR
DCA QR
JMP LOOP1
TAD L
CIA NU
SNA CLA
JMP I FFT
TAD SHFCHK
DCA SHFLAG
TAD SHFCHK
SNA CLA
ISZ SCAL
DCA SHFCHK
ISZ L
TAD S
CLL RAR
DCA S
ISZ S
NOP F
CMA
TAD N
TAD XRLOC
DCA PR
CLA IAC
DCA C
TAD PR
TAD XL0CDF
DCA PI
TAD XRLOC
CIA
TAD PR
DCA P
TAD F
SNA
JMP NOROT
CMA
TAD SHIFCT
DCA P
LSR
HLT
SKP
TAD
JMS I INVERT
TAD MNOVR2
JMS I GETRIG
NOP SINE
DCA I DB
TAD I DB

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0434 /DO REAL PART FIRST=RE[X(P)]*COSINE+I*IM[X(P)]*SINE
0435 /RE[X(P)]*COSINE=RE[X(P)]*RE[W*K]
0436 /C FOR ADDITION LATER
0437 /GET IM[X(P)]
0438
0439 /AC=IM[X(P)]*SINE=-IM[W*K]*IM[X(P)]
0440 /AC=RE[W*K]*RE[X(P)]-IM[W*K]*IM[X(P)]=RE[X(P)]*W*K
0441 /STORE AT GR
0442
0443 /DO IMAG, PART NEXT=IM[X(P)]*COSINE-RE[X(P)]*SINE=IM[X(P)]*RE[W*K]+RE[X(P)]*I*IM[W*K]
0444
0445 /AC=IM[X(P)]
0446
0447 /AC=IM[X(P)]*COSINE=IM[(P)]*RE[W*K]
0448 /STORE FOR LATER ADDITION
0449
0450 /AC=RE[X(P)]
0451
0452 /AC=RE[X(P)]*SINE=-RE[X(P)]*IM[W*K]
0453 /AC=RE[X(P)]*IM[W*K]
0454
0455 /AC=IM[X(P)]*RE[W*K]+RE[X(P)]*IM[W*K]=IM[X(P)]*W*K
0456 /STORE AT GI, SO GI=IM[X(P)]*W*K AND GR=RE[X(P)]*W*K G=GR+I*GI
0457 /LOCATE P NODE PAIR Q, LOCATED S=N/(2*L) UP ARRAY
0458 /SO SET Q=P-S=INDEX OF NODE PAIR
0459 /LOCATE X(Q) IN MEMORY BY FIXING POINTERS QR AND QI
0460 /TO QS REAL AND IMAG PARTS RESPECTIVELY
0461
0462 /DO THE COMPLEX OPERATIONS: X(P)<=X(Q)-G*X(Q)+X(Q)*G
0463 /FIRST DO REAL PART OF X(P), GET RE[X(Q)] AND STORE
0464 /GET RE[IG]
0465
0466 /SUBTRACT THEM,
0467 /RE[X(P)]<=RE[X(Q)]-RE[IG]
0468 /COMPUTE IMAG, PART OF X(P), GET IM[X(Q)]
0469 /AND STORE
0470 /GET IM[IG]
0471
0472 /AND SUBTRACT THEM,
0473 /IM[X(P)]<=IM[X(Q)]-IM[IG], X(P) IS NOW DONE.
0474 /NEXT COMPUTE X(Q), FIRST REAL PART
0475 /GET RE[IG] AND STORE
0476 /GET RE[IG] AND ADD TO FORM
0477 /RE[X(Q)]<=RE[X(Q)]+RE[IG]
0478 /RE[X(Q)]<=RE[X(Q)]+RE[IG]
0479 /NOW COMPUTE IMAG PART OF X(Q), GET IM[X(Q)]
0480 /AND STORE
0481 /GET IM[IG] AND ADD TO FORM
0482 /IM[X(Q)]<=IM[X(Q)]+IM[IG]
0483 /IM[X(Q)]<=IM[X(Q)]+IM[IG], THE NEW NODE PAIR IS COMPUTED.
0484 /MOVE UP ARRAY TO NEXT NODE, SET AC=-1
0485 /TO FORM -1
0486 /PC=P-1
0487
0488 /DO THE SAME FOR POINTER PR
0489
0490 /CHECK ON SPACING, IS A NODE WHICH HAS ALREADY BEEN COMPUTED
0491 /ABOUT TO BE RE-DONE, OR EQUIVALENTLY.
0492 /IS C=S?
0493 /YES.
0494 /NO, DO NEXT NODE PAIR
0495 /YES, BUT ARE WE AT THE TOP OF THE ARRAY?
0496 /OR, IS S=P+1? (P COMPLEMENTED=-P-1)
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0533 0671 5466 JMP I RECHK
0534 2672 1006 TAD S
0535 2673 7041 CIA
0536 0674 1025 TAD PR
0537 2675 3025 DCA PR
0540 2676 5465 JMP I RESETC
0541 2677 2032 ISZ C
0542 2700 5464 JMP I RBUILD
0543 2721 2020 SORTX, Q
0544 2702 7040 CMA
0545 2703 1003 TAD
0546 2704 3027 DCA Q
0547 2705 1027 TAD Q
0550 2706 4443 JMS I INVERT
0551 2707 3030 DCA P
0552 2710 1030 TAD P
0553 2711 7041 CIA
0554 2712 1027 TAD
0555 2713 7750 SPA SNA CLA
0556 2714 5345 JMP SWAPED
0557 2715 1030 TAD P
0560 2716 1051 TAD XRLOC
0561 2717 3025 DCA PR
0562 2720 1027 TAD Q
0563 2721 1051 TAD XRLOC
0564 2722 3023 DCA QR
0565 2723 1025 TAD PR
0566 2724 1052 TAD XLOCDF
0567 2725 3026 DCA PI
0570 2726 1023 TAD QR
0571 2727 1052 TAD XLOCDF
0572 2730 3024 DCA QI
0573 2731 1425 TAD I PR
0574 2732 3034 DCA TEMPR
0575 2733 1423 TAD I QR
0576 2734 3425 DCA I PR
0577 2735 1034 TAD TEMPR
0600 2736 3423 DCA I QR
0601 2737 1426 TAD I PI
0602 2740 3034 DCA TEMPR
0603 2741 1424 TAD I QI
0604 2742 3426 DCA I PI
0605 2743 1034 TAD TEMPR
0606 2744 3424 DCA I QI
0607 2745 1027 SWAPED, TAD Q
0610 2746 7640 SZA CLA
0611 2747 5352 JMP .+3
0612 2750 6201 CDF0
0613 2751 5701 JMP I SORTX
0614 2752 7040 CMA Q
0615 2753 1027 TAD Q
0616 2754 3027 DCA REVERS
0617 2755 5305 JMP
0620 EJECT

```

/YES, DONE WITH THIS ARRAY, DO NEXT ONE,
/NO, MOVE PAST ARRAY, THAT HAS ALREADY BEEN DONE, OR SET P TO P-S,
/BY CHANGING THE POINTER TO REIX(P)]

/REINITIALIZE C TO 1 SINCE AN UNUSED AREA HAS BEEN ENTERED.
/CK=C+1, ANOTHER NODE PAIR HAS BEEN HANDLED.
/DO NEXT NODE PAIR IN THIS AREA,
/SUBROUTINE THAT
/SORTS OUT TRANSFORMS BY
/BIT INVERSION OF ADDRESS,
/Q<=N-1, START FROM BOTTOM OF BUFFER
/P<=BIT INVERTED Q
/BIT INVERSION ROUTINE
/FORM Q-P

/IS PQ?
/NO, HAVE ALREADY DONE THIS PAIR
/YES, SWAP ORDER
/FIRST SET UP SUBSCRIPT POINTERS FOR X(P) AND X(Q),

/EXCHANGE: X(P)<=X(Q) AND X(Q)<=X(P)
/EXCHANGE REAL PARTS, GET REIX(P)]
/STORE IT,
/GET REIX(Q)]
/MAKE IT REIX(P)]
/GET REIX(P)]
/MAKE IT REIX(Q)]
/EXCHANGE IMAGINARY PARTS, GET IMCX(P)]
/STORE IT,
/GET IMCX(Q)]
/MAKE IT IMCX(P)]
/GET IMCX(P)]
/MAKE IT IMCX(Q)]
/IS Q=0?, IE; ARE WE AT THE TOP OF THE ARRAY

/YES, DONE EXIT
/NO, Q<=Q-1, IE; MOVE UP THE ARRAY

/GO BACK AND CONTINUE

```

0620 *1000
0621 /SIGNED S.P. MULTIPLY, USING THE
0622 /ENTRY: AC=MULTIPLIER, C(CALL+1)=ADDR OF MULTIPLICAND, EXIT=AC=PRODUCT,
0623 /AN 11 BIT SIGNED BINARY FRAC
0624 MULTIP, 0
0625 /AC=ARG1 (MULTIPLIER)
0626 /ARG1>0?
0627
0628 CML IAC
0629 SPA
0630 CMA CML IAC
0631 /NO-MAKE POS-SET L=1 TO SHOW IT WAS NEG
0632 /LOAD INTO M0
0633 >MOL
0634 CDF0
0635 TAD I
0636 /GET ADDR OF MULTIPLICAND
0637 DCA ARG2
0638 /STORE
0639 TAD I
0640 /AND RETRIEVE MULTIPLICAND ITSELF,
0641 /FOR EXIT AT CALL+2)
0642 ISZ MULTIP
0643
0644 SPA
0645 /ARG2>0?
0646 /NO, MAKE POSITIVE, CHANGE LINK, SINCE -1+--1=1 AND -1+1=-1
0647 /PUT AWAY AT ARG2
0648
0649 ARG2,
0650 /SIGN IN LINK, PUT INTO AC11 AND
0651 /PUT AWAY AT SIGN (=1 IF --1 =0 IF +)
0652 /DO MULTIPLICATION
0653 /ARGUMENT 2 (MULTIPLICAND)
0654 /NORMALIZE BINARY POINT,
0655 /SAVE HIGH ORDER, NOW ROUND OFF,
0656 /SET AC11=M00, AC0-10=0
0657
0658 SPA
0659 CLA CLL CMA RAR
0660 NOP
0661 SZL
0662 CMA IAC
0663 CDF1
0664 JMP I
0665 MULTIP
0666 SIGN,
0667 /BIT INVERSION ROUTINE
0668 /ENTRY: AC=WORD TO BE INVERTED; EXIT=AC=RESULT
0669 /NU CONTAINS THE NUME OF BITS IN THE WORD
0670 INVRT, 0
0671
0672 DCA WORD
0673 DCA WORDP
0674 NU
0675 CIA
0676 DCA FLIPCT
0677 TAD WORD
0678 CLL RAR
0679 DCA WORD
0680 TAD WORDP
0681 RAL
0682 DCA WORDP
0683 ISZ FLIPCT
0684 JMP FLIP
0685 TAD WORDP
0686 JMP I
0687 EJECT
0688
0689 1000 0000
0690 1001 7100
0691 1002 7510
0692 1003 7061
0693 1004 7421
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0695 1006 1600
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1252 1563 6201
1253 1564 1600
1254 1565 3220
1255 1566 1620
1256 1567 2200
1257 1568 7510
1258 1569 7061
1259 1570 3220
1260 1571 7421
1261 1572 6201
1262 1573 1600
1263 1574 3220
1264 1575 1620
1265 1576 2200
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1268 1579 3220
1269 1580 7421
1270 1581 6201
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1279 1590 6201
1280 1591 1600
1281 1592 3220
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1283 1594 2200
1284 1595 7510
1285 1596 7061
1286 1597 3220
1287 1598 7421
1288 1599 6201
1289 1600 1600

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Address	Instruction	Comment
0713	W/L3	
0714	/AND OF COS(2*PI*C(AC)/N) FOR C(AC) < N/2+1	
0715	/ENTRY: AC=INDEX OF LOOP UP	
0716	/EXIT: COS(2*PI*C(AC)/N) STORED AT "COSINE" AND	
0717	/AC=VALUE OF SIN(2*PI*C(AC)/N).	
0720	TRIGET, 0	
0721	0000	
0722	1061 6201	
0723	1062 3031	
0724	1063 7421	
0725	1064 1031	
0726	1065 7141	
0727	1066 1020	
0728	1067 3333	
0729	1070 7430	
0730	1071 5310	
0731	1072 1333	
0732	1073 7041	
0733	1074 7417	
0734	1075 0000	
0735	1076 7413	
0736	1077 7402	
0737	1100 1050	
0740	1101 3334	
0741	1102 1734	
0742	1103 7041	
0743	1104 3036	
0744	1105 1333	
0745	1106 1020	
0746	1107 5322	
0747	1110 1333	
0750	1111 7417	
0751	1112 0000	
0752	1113 7413	
0753	1114 7402	
0754	1115 1050	
0755	1116 3334	
0756	1117 1734	
0757	1120 3036	
0760	1121 1031	
0761	1122 7417	
0762	1123 0000	
0763	1124 7413	
0764	1125 7402	
0765	1126 1050	
0766	1127 3334	
0767	1130 1734	
0770	1131 6211	
0771	1132 5660	
0772	1133 0000	
0773	1134 0000	
0774		
0775		
0776		
0777		
1000		
1001		
1002		
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1008		
1009		
1010		

1012	1145	3374			
1013	1146	1033	DCA	ADD1	
1014	1147	7415	TAD	ADD2	
1015	1150	0000	ASR		
1016	1151	3033	0		
1017	1152	7501	DCA	ADD2	
1020	1153	7004	MQA		
1021	1154	7060	RAL		
1022			CMA CML		
1023	1155	7720	SMA SNL CLA		
1024	1156	7001	IAC		
1025	1157	1374	ADDWOS, TAD	ADD1	
1026	1160	1033	TAD	ADD2	
1027	1161	5375	DCA	XSUM	
1030	1162	1375	TAD	XSUM	
1031	1163	7510	SPA		
1032	1164	7041	CIA		
1033	1165	7004	RAL		
1034	1166	7700	SMA CLA	NOTNOR	
1035	1167	5372	JMP		
1036	1170	7001	IAC		
1037	1171	3055	DCA	SHFCHK	
1040	1172	1375	NOTNOR, TAD	XSUM	
1041	1173	5735	JMP I	ADDR	
1042	1174	0000	ADD1, 0		
1043	1175	0000	XSUM, 0		
1044			EJECT		

	/	=LO(ADD2)	
	/	MQ(1)=LO(ADD(1))	
	/	GET MQ	
	/	L<=LO(ADD2); AC0<=LO(ADD1)	
	/	COMPLEMENT BOTH,	
	/	IF BOTH WERE=1 (NEITHER=0), INTRODUCE A CARRY.	
	/	DO THE ADDITION,	
	/	STORE THE RESULT	
	/	CHECK TO SEE IF ALREADY NORMALIZED,	
	/	IS IT POSITIVE?	
	/	MAKE IT POSITIVE.	
	/	GET BIT 1, WAS NORMALIZED IF #1	
	/	NOT NORMALIZED, LEAVE SHFCHK ALONE,	
	/	SET SHFCHK=1	
	/	AND EXIT	
	/	ADDEND STORAGE	
	/	TEMP STORAGE FOR SUM	

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/DEFINITIONS FOR EAE
DVI=7407
NMI=7411
SHL=7413
ASR=7415
LSR=7417
MQL=7421
MUY=7405
MOA=7501
CAM=7621
SCA=7441
SCL=7403
/ASSEMBLY PARAMETERS
BIGSNU=12 /LARGEST TRANSFORMATION HAS DIMENSION 2*10.
EJECT

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1065 /MOVING WINDOW DISPLAY SUBROUTINE
1070 PMODE
1071 PAGE
1072 0
1073 /GET BOUNDS
1074 IDORA, 0000
1075 1200 0000
1076 1201 7300
1077 1202 6201
1078 1203 1600
1079 1204 3635
1080 1205 2200
1081 1206 1600
1082 1207 3636
1083 1210 2200
1084 1211 1600
1085 1212 3637
1086 1213 2200
1087 1214 7001
1088 1215 1600
1089 1216 3640
1090 1217 7004
1091 1220 1637
1092 1221 3637
1093 1222 2200
1094 1223 1600
1095 1224 3111
1096 1225 2200
1097 1226 1600
1098 1227 3536
1099 1230 1635
1100 1231 3641
1101 1232 1636
1102 1233 3642
1103 1234 5600
1104 1235 1415
1105 1236 1416
1106 1237 1474
1107 1240 1475
1108 1241 1574
1109 1242 1575
1110 1243 0401
1111 1244 1243
1112 1245 3274
1113 1246 1112
1114 1247 4261
1115 1250 1113
1116 1251 4261
1117 1252 1114
1118 1253 4261
1119 1254 1115
1120 1255 1243
1121 1256 4261
1122 1257 0000
1123 1260 5743
1124 1261 0000
1125 1262 6141
1126 1263 5276
1127 1264 4001
1128 1265 0024
1129 1266 0265
1130 1267 1020
1131 1270 7757
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/GET BOUNDS
/ DATA BUFFER
/15 BIT
/LOWER BOUND
/AT P+1, P+2
/MINFLD,MINADR
/UPPER BOUND
/AT P+3, P+4
/RDORA USES
/MAX+1
/Y SHIFT
/Y SCALE
/INITIALIZE
/WINDOW
/STARTING ADDR
/RTN TO SCR N
/DSC X,Y COORD
/FIELD
/ADDRESS
/CONTENTS OF
/CURSR CORE LOC
/Y COORD OF
/CURSOR POINT
/RESTORE USER
/ DATA FLD
/RTN
/DSC C(AC)
/SAVE VALUE
/CHAN 1
/VC FOR FULL
/ SIZE IS -40
/-20 FOR HALF

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1362	1452	JMP OKEND	/RESET TO
1363	1456	TAD MINADR	/LOWER BOUND
1364	1457	DCA BUFPTR	
1365	1460	TAD MINFLD	
1366	1461	DCA BOUND	
1367	1462	JMP NXTDF	
1370	1463	ISZ BUFPTR	/CHK FOR FIELD
1371			/BOUNDARY
1372	1464	JMP OKFLD	/ITS OK
1373	1465	ISZ BOUND	/SET NXT FLD
1374	1466	JMS SETDF	
1375	1467	ISZ COUNT	/512 PNTS ?
1376	1470	JMP NXTPTNT	/NO
1377	1471	JMP I ,+1	/DSC READ OUT
1400	1472	DSCLOC	
1401	1473	JMS BOUND	/CHK UPR BOUND
1402	1474	MAXFLD, 2	
1403	1475	MAXADR, 0	
1404	1476	SPA CLA	/HI WRAP ?
1405	1477	JMP SETFLD	/YES
1406	1500	TAD MINFLD	/RESET TO
1407	1501	DCA BUFHI	/LOWER BOUND
1410	1502	TAD MINADR	
1411	1503	JMP WRAP	
1412		/DOUBLE PRECISION ADD	
1413		/((DBLHI,DBLLO)*((BUFHI,BUFLO)	
1414		/RESULT IN (DBLHI,DBLLO)	
1415		/((BUFHI,BUFLO)=INITIAL SCOPE ADDRESS	
1416			
1417	1504	DADD, 0	
1420	1505	CLA CLL	
1421	1506	TAD DBLLO	
1422	1507	TAD BUFLO	
1423	1510	DCA DBLLO	
1424	1511	RAL	
1425	1512	TAD DBLHI	
1426	1513	TAD BUFHI	
1427	1514	DCA DBLHI	
1430	1515	JMP I DADD	
1431			
1432		/ADD -UPPER OR -LOWER BOUND	
1433		/TO (BUFHI,BUFLO)	
1434		/BOUND IS AT P+1,P+2 OF CALL	
1435			
1436	1516	BOUND, 0	
1437	1517	TAD I BOUND	/2S COM OF ARG
1440	1520	CMA CLL	/TO DAC
1441	1521	DCA DBLHI	
1442	1522	ISZ BOUND	
1443	1523	TAD I BOUND	
1444	1524	CIA	
1445	1525	SZL	
1446	1526	ISZ DBLHI	
1447	1527	NOP	
1450	1530	M1000,	
1451	1531	DCA DBLLO	
1452	1532	JMS DADD	
1453	1533	TAD DBLHI	
1454	1534	DCA ENDHI	/DAC HOLDS -NUM
1455	1535	TAD DBLLO	/TO END OF BUF
1456	1536	DCA ENDLO	/NO MATTER F
1457	1537	TAD DBLHI	/LOW END WRA
			/TO CHK FOR

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/SET 8 F
/REL TO 0

JMP I BOUND
0
TAD BOUND
CLL RTL
RAL
TAD CCDF0
DCA ,+1
0
JMP I SETDF
CURDIS, DCA YCUR
TAD BOUND
DCA XCURHI
TAD BUFPTR
DCA XCURL0
TAD I BUFPTR
DCA CORVAL
TAD M70
DCA DBLLO
TAD YCUR
LINC
LMODE
SNS I 5
JMP FREE
DIS XCORD
POP
PMODE
ISZ DBLLO
JMP CURLOP
JMP CURRTN
0
CURCNT, 0
/THESE 5 GUYS MAY BE PAGE 0
BUFHI, 1
BUFLO, 0
ENDLO, 0
ENDHI, 0
DBLHI=SETDF
BUFPTR=DADD
XCORD=1
LMODE
CURSAM=SAM 1
WINSAM=SAM 0
FRESAM=SAM 5
SCALE=SCR
SC12BU=SCR 3
OF12BU=4000
CHAIN "FFTC-2"

/DISP CURSOR
/SAVE X,Y
/COORDINATES

/CURSOR KNOB
/WINDOW KNOB
/FREE CURSOR

/SCALE FACTOR
/12 BIT UNSIGNED
/Y OFFSET FOR
/12 BIT UNSIGNED

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0000
0001
EJECT

LINE	ADDRESS	DATA	INSTR	COMMENT
0002	0000	0000	LMODE	
0003	0001	0000	SEGMENT 2	
0004	0002	0000	*20	
0005	0003	0000	LDF	7 /BOOTSTRAP IN DIAL MS I/O ROUTINES
0006	0004	0000	RDC	
0007	0005	0000	6322	
0008	0006	0000	RDC	
0009	0007	0000	7323	
0010	0008	0000	LDF	3
0011	0009	0000	LDA I	/INPUT FROM DIAL TAPE?
0012	0010	0000	QUES1+2000	
0013	0011	0000	LIF 2	
0014	0012	0000	JMP	ASK
0015	0013	0000	LJ4	
0016	0014	0000	ANSWER+6000	
0017	0015	0000	SAE I	
0018	0016	0000	31	
0019	0017	0000	SKP	
0020	0018	0000	JMP	UNTFIL /DIAL
0021	0019	0000	SAE I	
0022	0020	0000	16	
0023	0021	0000	JMP	IFDIAL /ERROR
0024	0022	0000	JMP	DATTAP
0025	0023	0000	ASK FOR UNIT NO + FILE NAME	
0026	0024	0000	UNTFIL, JMP	ASK2
0027	0025	0000	JMP	:-1
0028	0026	0000	LJF	1
0029	0027	0000	LDA I	
0030	0028	0000	FDV+2000	
0031	0029	0000	20	
0032	0030	0000	SKP	
0033	0031	0000	JMP	MOVINP
0034	0032	0000	LJF	2
0035	0033	0000	LDA I	
0036	0034	0000	MSG1+2000	
0037	0035	0000	JMP	ASK
0038	0036	0000	JMP	UNTFIL
0039	0037	0000	JMP	ASK3
0040	0038	0000	JMP	DATTAP
0041	0039	0000	MOVINP, JMP	FDV2RW
0042	0040	0000	PTS,	3
0043	0041	0000	LJF	2
0044	0042	0000	LDA I	
0045	0043	0000	QUES4+2000	
0046	0044	0000	JMP	ASK
0047	0045	0000	JMP	ASK
0048	0046	0000	JMP	ASK
0049	0047	0000	JMP	ASK
0050	0048	0000	JMP	ASK
0051	0049	0000	JMP	ASK
0052	0050	0000	JMP	ASK
0053	0051	0000	JMP	ASK
0054	0052	0000	JMP	ASK
0055	0053	0000	JMP	ASK
0056	0054	0000	JMP	ASK
0057	0055	0000	JMP	ASK
0058	0056	0000	JMP	ASK
0059	0057	0000	JMP	ASK
0060	0058	0000	JMP	ASK
0061	0059	0000	JMP	ASK
0062	0060	0000	JMP	ASK
0063	0061	0000	JMP	ASK
0064	0062	0000	JMP	ASK
0065	0063	0000	JMP	ASK
0066	0064	0000	JMP	ASK
0067	0065	0000	JMP	ASK
0068	0066	0000	JMP	ASK
0069	0067	0000	JMP	ASK
0070	0068	0000	JMP	ASK
0071	0069	0000	JMP	ASK
0072	0070	0000	JMP	ASK
0073	0071	0000	JMP	ASK
0074	0072	0000	JMP	ASK
0075	0073	0000	JMP	ASK
0076	0074	0000	JMP	ASK
0077	0075	0000	JMP	ASK
0078	0076	0000	JMP	ASK
0079	0077	0000	JMP	ASK

[illegible]

0200	0212	2407	RWPARM+2002	
0201	0213	1100	ADA	
0202	0214	2410	RWPARM+2003	
0203	0215	1120	ADA I	
0204	0216	6777	-1000	
0205	0217	0471	APD I	PTS
0206	0220	6064	JMP	/YES
0207	0221	0002	PDP	
0210			PMODE	
0211	4222	7200	CLA	
0212	4223	1003	TAD N /ADD 1 BLK FOR SCALE FACTOR IF 400 WORDS OR MORE	
0213	4224	7104	CLL RAL	/NO OF OUTPUT WRDS = NO OF PTS*2
0214	4225	1107	TAD M400	
0215	4226	7700	SMA CLA	
0216	4227	2523	ISZ I PFDV7	
0217	4230	6141	LINC	
0220			LMODE	
0221	0231	1020	LDA I	/DO FFT OR JUST DISPLAY?
0222	0232	2625	QUES11+2000	
0223	0233	6720	JMP ASK	
0224	0234	1300	LDH	
0225	0235	7043	ANSWER+6000	
0226	0236	1460	SAE I	
0227	0237	0004	4	
0230	0240	6244	JMP .+4	
0231	0241	1060	STA I	/NOT=0 JUST DISPLAY
0232	0242	0000	DISFLG, 0	
0233	0243	6251	JMP FIF	
0234	0244	1460	SAE I	
0235	0245	0006	6	
0236	0246	6231	JMP IFFFT	/ERROR
0237	0247	0011	CLR	
0240	0250	4242	STC	DISFLG
0241	0251	1300	LDH	/=0 WILL DO TRANSFORM OR INVERSE
0242	0252	7044	ANSWER+6001	
0243	0253	1460	SAE I	
0244	0254	0024	24	
0245	0255	6261	JMP IFI	
0246	0256	0011	CLR	
0247	0257	4356	STC IFTFLG	/DO FFT
0250	0260	6265	JMP IFDISP	
0251	0261	1460	SAE I	
0252	0262	0011	11	
0253	0263	6231	JMP IFFFT	
0254	0264	4356	STC IFTFLG	/DO IFFT
0255	0265	2242	IFDISP, ADD	DISFLG
0256	0266	0470	AZE I	
0257	0267	6273	JMP	OUTQES
0260	0270	6466	JMP	F0V2RW
0261	0271	0603	LIF	3
0262	0272	6001	JMP	DISPLY
0263			/GET OUTPUT INFO	
0264	0273	1020	OUTQES, LDA I	
0265	0274	2571	QUES5+2000	
0266	0275	6720	JMP	ASK
0267	0276	1300	LDH	
0270	0277	7043	ANSWER+6000	
0271	0300	1460	SAE I	
0272	0301	0031	31	
0273	0302	0456	SKP	
0274	0303	6310	JMP	OUTUNT
0275	0304	1460	SAE I	

/PUT ON DIAL TAPE?

0277	0300	0301	0302	0303	0304	0305	0306	0307	0310	0311	0312	0313	0314	0315	0316	0317	0320	0321	0322	0323	0324	0325	0326	0327	0330	0331	0332	0333	0334	0335	0336	0337	0340	0341	0342	0343	0344	0345	0346	0347	0350	0351	0352	0353	0354	0355	0357	0360	0361	0362	0363	0364	0365	0366	0367	0370	0371	0372	0373	0374	0375	0376	0377	0378	0379	0380	0381	0382	0383	0384	0385	0386	0387	0388	0389	0390	0391	0392	0393	0394	0395	0396	0397	0398	0399	0400	0401	0402	0403	0404	0405	0406	0407	0408	0409	0410	0411	0412	0413	0414	0415	0416	0417	0418	0419	0420	0421	0422	0423	0424	0425	0426	0427	0428	0429	0430	0431	0432	0433	0434	0435	0436	0437	0438	0439	0440	0441	0442	0443	0444	0445	0446	0447	0448	0449	0450	0451	0452	0453	0454	0455	0456	0457	0458	0459	0460	0461	0462	0463	0464	0465	0466	0467	0468	0469	0470	0471	0472	0473	0474	0475	0476	0477	0478	0479	0480	0481	0482	0483	0484	0485	0486	0487	0488	0489	0490	0491	0492	0493	0494	0495	0496	0497	0498	0499	0500	0501	0502	0503	0504	0505	0506	0507	0508	0509	0510	0511	0512	0513	0514	0515	0516	0517	0518	0519	0520	0521	0522	0523	0524	0525	0526	0527	0528	0529	0530	0531	0532	0533	0534	0535	0536	0537	0538	0539	0540	0541	0542	0543	0544	0545	0546	0547	0548	0549	0550	0551	0552	0553	0554	0555	0556	0557	0558	0559	0560	0561	0562	0563	0564	0565	0566	0567	0568	0569	0570	0571	0572	0573	0574	0575	0576	0577	0578	0579	0580	0581	0582	0583	0584	0585	0586	0587	0588	0589	0590	0591	0592	0593	0594	0595	0596	0597	0598	0599	0600	0601	0602	0603	0604	0605	0606	0607	0608	0609	0610	0611	0612	0613	0614	0615	0616	0617	0618	0619	0620	0621	0622	0623	0624	0625	0626	0627	0628	0629	0630	0631	0632	0633	0634	0635	0636	0637	0638	0639	0640	0641	0642	0643	0644	0645	0646	0647	0648	0649	0650	0651	0652	0653	0654	0655	0656	0657	0658	0659	0660	0661	0662	0663	0664	0665	0666	0667	0668	0669	0670	0671	0672	0673	0674	0675	0676	0677	0678	0679	0680	0681	0682	0683	0684	0685	0686	0687	0688	0689	0690	0691	0692	0693	0694	0695	0696	0697	0698	0699	0700	0701	0702	0703	0704	0705	0706	0707	0708	0709	0710	0711	0712	0713	0714	0715	0716	0717	0718	0719	0720	0721	0722	0723	0724	0725	0726	0727	0728	0729	0730	0731	0732	0733	0734	0735	0736	0737	0738	0739	0740	0741	0742	0743	0744	0745	0746	0747	0748	0749	0750	0751	0752	0753	0754	0755	0756	0757	0758	0759	0760	0761	0762	0763	0764	0765	0766	0767</
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0376	4402	7040	CMA	/NEW ADPP	NO OF PTS
0	4403	1003	TAD		
0	4404	3011	DCA		
0401	4405	1003	TAD		
0402	4406	7041	CIA		
0403	4407	3034	DCA	TEMPR	/CTR
0404	4410	3130	DCA	CMPLG	/DONT COMPLEMENT
0405	4411	4527	JMS I	PMVPTS	/PACK IMAG PARTS BEHIND REAL
0406	4412	6141	LINC		
0407			LMODE		
0410	0413	6466	JMP FDV2RW		
0411	0414	0002	PDP		
0412			PMODE		
0413	4415	6212	CIF		/WRITE OUT DATA
0414	4416	4535	JMS I	10	
0415	4417	6405	RWPARM	PWRITE	
0416	4420	6141	LINC		
0417			LMODE		
0420	0421	0603	LIF 3		
0421	0422	6001	JMP DISPLY		
0422	0423	0602	LIF 2		
0423	0424	1020	LDA I		
0424	0425	3013	MSG2+2000		
0425	0426	6720	JMP	ASK	
0426	0427	6273	JMP	OUTQES	/ASK OUTPUT QUESTIONS AGAIN
0427	0430	0602	LIF 2		
0430	0431	1020	LDA I		/NAME ALREADY EXISTS
0431	0432	2612	QUES6+2000		
0432					/REPLACE WITH NEW FILE?
0433	0433	6720	JMP	ASK	
0434	0434	1300	LDH		
0435	0435	7043	ANSWER+6000		
0436	0436	1460	SAE I		
0437	0437	0031	31		
0440	0440	0456	SKP		
0441	0441	6446	JMP	REPL	
0442	0442	1460	SAE I		
0443	0443	0016	16		
0444	0444	6430	JMP	SAMNAM	
0445	0445	6310	JMP	OUTUNT	
0446	0446	0601	LIF 1		
0447	0447	6024	JMP	24	
0450	0450	6423	JMP	NOSPAC	
0451	0451	6320	JMP	RDDATA	
0452	0452	0602	LIF 2		
0453	0453	6572	JMP	ASK3	/ASK FOR UNIT/BLK NO
0454	0454	6452	JMP	ONDAT	/ERROR
0455	0455	1000	LDA		/BLK NO
0456	0456	2403	FDV+2006		
0457	0457	1100	ADA		/NO OF BLKS
0460	0460	2404	FDV+2007		
0461	0461	1120	ADA I		
0462	0462	6777	-1000		
0463	0463	0471	APO I		
0464	0464	6423	JMP	NOSPAC	/NOT ENOUGH BLKS LEFT
0465	0465	6320	JMP	RDDATA	
0466				/MOVE FDV PARAMETERS TO R-W LIST	
0467	0466	1000	FDV2RW,	LDA	
0470	0467	2375		FDV+2000	
0471	0470	1040		STA	
0472	0471	2405		RWPARM+2000	

0476	0475	1040	SIA	
0477	0476	2407	RWPARM+2002	
0500	0477	1000	LDA	
0501	0477	2404	FDV+2007	
0502	0501	1040	STA	
0503	0501	2410	RWPARM+2003	
0504	0502	6000	JMP 0	
0505	0503	4114	/CONVERT WORDS TO BLOCKS	
0510	0504	2000	NUMBKX, STC TEMP1	
0511	0505	4522	ADD 0	
0512	0506	2114	STC NUMBKX	
0513	0507	0643	ADD TEMP1	
0514	0508	0061	LDF 3	
0515	0509	0001	SET I 1	
0516	0510	1120	ADA I	
0517	0511	7377	-400	
0518	0512	0451	AP0	
0519	0513	6520	JMP ,+3	
0520	0514	0221	XSK I 1	
0521	0515	6512	JMP ,+5	
0522	0516	1000	LDA	
0523	0517	0001	1	
0524	0518	0000	NUMBKX, 0	
0525	0519		/ASK FOR UNIT NUMBER & FILE NAME	
0526	0520		/CONV & STORE UNIT NUMBER	
0527	0521		/MOVE FILE NAME TO ENTER, LOOKUP PARAMETER LIST	
0530	0522		/STORE UNIT THRU B3	
0531	0523	1000	ASK2, LDA	
0532	0524	0000	0	
0533	0525	4571	STC ASK2X	
0534	0526	0602	LIF 2	
0535	0527	6711	JMP OCTL	/CHANGE PARAMETERS TO HANDLE OCTAL NUMBERS
0536	0528	1020	LDA I	
0537	0529	2453	QUES2+2000	
0540	0530	6720	JMP ASK	/PT TO UNIT NO-1H
0541	0531	0061	SET I 1	
0542	0532	3043	ANSWER+2000	
0543	0533	1020	LDA I	/MAX VALUE
0544	0534	3017	17	
0545	0535	5627	JMP CONV	
0546	0536	6571	JMP ASK2X	/ERROR
0547	0537	1040	STA	/STORE UNIT
0550	0538	2375	FDV+2000	
0551	0539	0061	SET I 1	
0552	0540	7044	ANSWER+6001	
0553	0541	0062	SET I 2	
0554	0542	6375	FDV+6000	/LEFT HALF 1ST OF FDV+1
0555	0543	0063	SET I 3	
0556	0544	7767	-10	/8 CHARS
0557	0545	1321	LDH I 1	/IF 1ST CHAR OF NAME
0560	0546	0470	AZE I	/=00, NO NAME WAS
0561	0547	6571	JMP ASK2X	/ENTERED-ERROR
0562	0548	0456	SKP	
0563	0549	1321	LDH I 1	
0564	0550	0450	AZE	/FILL TO 8 CHARS WITH 77
0565	0551	6562	JMP ,+3	
0566	0552	1320	LDH I	
0567	0553	7700	LDH I	
0570	0554	1362	STH I 2	
0571	0555	0223	XSK I 3	
0572	0556	4555	INB	INFILE

05	0565	1020	LDA I	/	IRN CALL+2
06	0566	0001	1		
07	0567	1140	ADM		
08	0570	0571	+1		
09	0571	0000	ASK2X, 2		
10			/ASK FOR UNIT NUMBER + BLK NO AND CONVERT		
11			/STORE UNIT THRU B7		
12			/ " BLK NO " B10		
13			ASK3, LDA		
14	0572	1000	2		
15	0573	0000	STC	ASK3X	
16	0574	4626	LIF	2	
17	0575	0602	LDA I		
18	0576	1020	QUES3+2000		
19	0577	2475	JMP	ASK	
20	0600	6720	SET I	1	
21	0601	0061	ANSWER+2000		
22	0602	3043	JMP OCTL		
23	0603	6711	LDA I		
24	0604	1020	17		
25	0605	0017	JMP CONV		
26	0606	6627	JMP ASK3X		
27	0607	6626	STA		
28	0610	1040	FDV+2000		
29	0611	2375	SET I	1	
30	0612	0061	ANSWER+6001		
31	0613	7044	LDA I		
32	0614	1020	777		
33	0615	0777	JMP	CONV	
34	0616	6627	JMP	ASK3X	
35	0617	6626	STA		
36	0620	1040	FDV+2006		
37	0621	2403	LDA I		
38	0622	1020	1		
39	0623	0001	ADM		
40	0624	1140	+1		
41	0625	0626	0		
42	0626	0000	ASK3X,		
43			/CONVERT NUMBER IN ANSWER BUFFER TO BINARY		
44			/ENTER WITH MAX LEGAL VALUE IN AC		
45			/IF LEGAL - EXIT CALL+2 WITH VALUE IN AC		
46	0627	0017	CONV,		
47	0630	4675	TEMP2 /COMPLEMENT MAX VALUE		
48	0631	4114	TEMP1		
49	0632	2000	0		
50	0633	4674	CONV		
51	0634	1321	NXTCHR, LDH I	1	
52	0635	0470	AZE I		
53	0636	6660	JMP ERRCHK		
54	0637	1120	ADA I		
55	0640	7720	7720		
56			/S COMP		
57	0641	0451	APD		
58	0642	6650	JMP	CHKEND	
59	0643	1301	LDH	1	
60	0644	1120	ADA I		
61	0645	7710	UPLEGL, -67		
62	0646	0451	APD		
63	0647	6676	JMP	MULPLY	
64	0650	1301	LDH	1	
65	0651	1460	SAE I		
66	0652	0000	14		

0674	0674	CONVER	ILLEGAL CHAR
0656	0674		/=34 OR 74 - NUMBER COMPLETED
0657	0674		
0660	1000	ERRCHK, LDA	
0661	0114	TEMP1	
0662	1100	ADA	
0663	0675	TEMP2	
0664	0471	AP0 I	
0665	0674	CONVER	
0666	1020	LDA I	
0667	0001	1	
0670	1140	ADM	
0671	0674	ADM	
0672	1000	ADM	
0673	0114	TEMP1	
0674	0000	CONVER, 0	
0675	0000	TEMP2, 0	
0676	1000	MULPLY, LDA	
0677	0114	TEMP1	
0700	1260	MUL I	
0701	0010	MPLIER, 10	
0702	4114	TEMP1	
0703	1301	STC	
0704	1560	LDH	
0705	7760	BCL I	
0706	1140	ADM	
0707	0114	TEMP1	
0710	6634	JMP	
0711		NXTCHR	
0712			
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[illegible]

1067	6070	3011	DCA	11		
1070	6071	1033	TAD	ADD2		
1071	6072	3034	UCA	TEMPR	/MOVE 1/2 OF PTS	
1072	6073	7201	CLA	IAC		
1073	6074	3130	DCA	CMPLG	/COMPLEMENT VALUES	
1074	6075	4527	JMS	I	/MOVE THEM	
1075	6076	7040	CMA		/OLD ADDR OF 2ND 1/2 = 3/2 NO OF PTS	
1076	6077	1037	TAD	GR		
1077	6100	1003	TAD	N		
1100	6101	3010	DCA	10		
1101	6102	1110	TAD	C1777	/NEW ADDR OF 2ND 1/2 = 2000	
1102	6103	3011	DCA	11		
1103	6104	1033	TAD	ADD2		
1104	6105	3034	DCA	TEMPR	/1/2 OF PTS	
1105	6106	4527	JMS	I	/MOVE THEM - 1ST 1/2 IS NOW 2ND 1/2; 2ND 1/2 IS NOW 1ST 1/2	
1106	6107	5330	JMP	PREPAR		
1107	6110	1003	TAD	N	/LOW ADDR OF IMAG = NO OF PTS	
1110	6111	3364	DCA	LOADDR	/HIGH ADDR = 2*NO OF PTS-1	
1111	6112	7040	CMA			
1112	6113	1003	TAD	N		
1113	6114	1003	TAD	N		
1114	6115	3366	DCA	UPADDR		
1115	6116	5330	JMP	PREPAR		
1116						
1117	6117	1533	DPREAL, TAD	I	PIFTFG	/IF TRANSFORM WAS DONE, SWAP HALVES
1120	6120	7640	SZA	CLA		
1121	6121	5324	JMP	I	NOSWPR	
1122	6122	4526	JMS	I	PMRLMG	/SWAP
1123	6123	5330	JMP	PREPAR		
1124	6124	3364	DCA	LOADDR	/LOW ADDR OF REAL	
1125	6125	7040	CMA			
1126	6126	1003	TAD	N		
1127	6127	3366	DCA	UPADDR	/HIGH ADDR = NO OF PTS-1	
1130						
1131	6130	1033	PREPAR, TAD	ADD2		
1132	6131	7104	CLL	RAL	/NO OF PTS <1000?	
1133	6132	1137	TAD	C1000		
1134	6133	7550	SPA	SNA		
1135	6134	5347	JMP	G01000		
1136	6135	7110	CLL	RAR	/YES	
1137	6136	7001	IAC		/CENTER DISPLAY	
1140	6137	1141	TAD	M1K		
1141	6140	3525	DCA	I	PLEFTX	/1000-(1000-NO OF PTS/2) 1,S COMP
1142	6141	1146	TAD	CCLR		
1143	6142	3524	DCA	I	PMVDIS	/WIDTH OF DISPLAY
1144	6143	1033	TAD	ADD2	/NO OF PTS	
1145	6144	7104	CLL	RAL		
1146	6145	3131	DCA	MINPTS		
1147	6146	5362	JMP	SHOWIT		
1150	6147	7200	CLA			
1151	6150	1141	TAD	M1K	/LEFT JUSTIFY DISPLAY	
1152	6151	5525	DCA	I	PLEFTX	/-1000 1,S COMP
1153	6152	1525	TAD	I	PLEFTX	/WIDTH OF DISPLAY
1154	6153	7001	IAC			
1155	6154	3131	DCA	MINPTS		
1156	6155	1145	TAD	SCR4		
1157	6156	3524	DCA	I	PMVDIS	/MOVE DISPLAY
1160	6157	5362	JMP	SHOWIT		
1161			/DISPLAY DATA			
1162	6160	6141	REDPLY, LINC			
1163			LMODE			
1164	0161	6001	JMP	DISPLY		

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1166
1167 * SHOWIT, JMS I KIDORA
1168 6162 4517 1
1169 6163 0001 /LOW ADD .ELD
1170 6164 0000 /" "
1171 6165 0001 /HIGH " "
1172 6166 0000 /" "
1173 6167 0000 /Y OFFSET
1174 6167 0000 /SCALE
1175 2170 0345 LMODE
1176 1170 SCR 3
1177 1177 PMODE
1200 6171 4520 JMS I KRORA
1201 6172 0331 KSF
1202 6173 5371 JMP -2
1203 6174 0336 KRB
1204 6175 1102 TAD M215
1205 6176 7650 SNA CLA
1206 6177 5360 JMP REDPLY
1207 6200 0036 KRB
1210 6201 1106 TAD M261
1211 6202 7650 SNA CLA
1212 6203 5216 JMP LARGER
1213 6204 0036 KRB
1214 6205 1103 TAD M321
1215 6206 7650 SNA CLA
1216 6207 5211 JMP SMALLR
1217 6210 5522 JMP I PRFRSH
1220 6211 1536 SMALLR, TAD I KYSCAL
1221 6212 1104 TAD M353
1222 6213 7710 SPA CLA
1223 6214 2536 ISZ I KYSCAL
1224 6215 5522 JMP I PRFRSH
1225 6216 1536 LARGER, TAD I KYSCAL
1226 6217 1105 TAD M340
1227 6220 7750 SPA SNA CLA
1230 6221 5522 JMP I PRFRSH
1231 6222 7040 CMA
1232 6223 1536 TAD I KYSCAL
1233 6224 3536 DCA I KYSCAL
1234 6225 5522 JMP I PRFRSH
1235
1236 /DISPLAY SCALE FACTOR
1237
1240 DPSCAL, TAD I PRELFG
1241 6226 1532 SZA CLA
1242 6227 7640 JMP I POSPER
1243 6230 5772 TAD N
1244 6231 1003 CLL RAL
1245 6232 7104 DCA TEMPR
1246 6233 3034 CDF1
1247 6234 6211 TAD I TEMPR
1250 6235 1434 TAD M11
1251 6236 1077 SMA SZA CLA
1252 6237 7740 JMP GR9
1253 6240 5244 TAD LESS10
1254 6241 1074 TAD I TEMPR
1255 6242 1434 JMP SHOSCL
1256 6243 5247 TAD M12
1257 6244 1071 GR9, TAD I TEMPR
1258 6245 1434 TAD GRET10
1259 6246 1073 TAD
1260 6247 6201 SHOSCL, CDF0
1261 6248 3266 DCA
1262 6250 3266 DPMAG-2 /STORE IN DISPLAY PARAMETERS
1263 6251 6141 I INC

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LINE	ADDRESS	DATA	INSTR	COMMENT
1200	0000	LDA 1		
1201	0001	SOLFAC+2000		
1202	0002	LIF 2		
1203	0003	JMP	ASK	/DISPLAY IT
1204	0004	JMP	DISPLY	
1205	0005	SOLFAC, TEXT Z		
1206	0006			
1207	0007			
1208	0008			
1209	0009			
1210	0010			
1211	0011			
1212	0012			
1213	0013			
1214	0014			
1215	0015			
1216	0016			
1217	0017			
1218	0018			
1219	0019			
1220	0020			
1221	0021			
1222	0022			
1223	0023			
1224	0024			
1225	0025			
1226	0026			
1227	0027			
1228	0028			
1229	0029			
1230	0030			
1231	0031			
1232	0032			
1233	0033			
1234	0034			
1235	0035			
1236	0036			
1237	0037			
1238	0038			
1239	0039			
1240	0040			
1241	0041			
1242	0042			
1243	0043			
1244	0044			
1245	0045			
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1250	0050			
1251	0051			
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1254	0054			
1255	0055			
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1259	0059			
1260	0060			
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1265	0065			
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1270	0070			
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1275	0075			
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1280	0080			
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1290	0090			
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1299	0099			
1300	0100			
1301	0101			
1302	0102			
1303	0103			
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1306	0106			
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1312	0112			
1313	0113			
1314	0114			

1352	0335	1240	MUL				
1353	0336	0000	IMGPTR, 0				
1354	0337	0363	SCR I	3	/SAME REASON	REAL	
1355	0340	0002	PDP				
1356			PMODE				
1357	6341	3035	DCA	SINE	/SAVE H.O.		
1360	6342	7100	CLL				
1361	6343	7501	MQA		/L.O. OF IMAG		
1362	6344	1142	TAD	DPSQ	/L.O. OF REAL		
1363	6345	3142	DCA	DPSQ			
1364	6346	7004	RAL		/OVERFLOW IF ANY		
1365	6347	1035	TAD	SINE	/H.O. OF IMAG		
1366	6350	1143	TAD	DPSQ+1	/H.O. OF REAL		
1367	6351	3143	DCA	DPSQ+1			
1370	6352	4773	JMS I	PSQRT	/TAKE SQ RT		
1371	6353	3436	DCA I	COSINE	/STORE IN PLACE OF REAL		
1372	6354	2036	ISZ	COSINE	/STEP REAL PTR		
1373	6355	2336	ISZ	IMGPTR	/ " IMAG ADDR - ON LAST PT OF 1024 PTS WILL SKIP		
1374	6356	2324	ISZ	RELPT	/STEP REAL ADDR		
1375	6357	2034	ISZ	TEMPR	/STEP CTR		
1376	6360	5320	JMP	NXTMAG			
1377	6361	6141	LINC				
1400			LMODE				
1401	0362	0643	LDF	3			
1402	0363	0002	PDP				
1403			PMODE				
1404	6364	1533	TAD I	PIFTG	/IF TRANSFORM WAS DONE, SWAP HALVES		
1405	6365	7640	SEA CLA				
1406	6366	5774	JMP I	PNSWPR			
1407	6367	4526	JMS I	PMRLMG			
1410	6370	5771	JMP I	PPREPR	/DISPLAY MAG		
1411			/				
1412	6371	6130	PPREPR,	PREPAR			
1413	6372	6051	PDSPER,	DISPER			
1414	6373	7052	PSQRT,	SORT			
1415	6374	6124	PNSWPR,	NOSWPR			
1416			EJECT				

1420	1420	LOOKUP, ENTER PARAMETER LIST	LMODE
1421	0375 0000	FDV, 0	/UNIT 0
1422	0376 0000	0	/FILE NAME - 8 CHAR
1423	0377 0000	0	
1424	0400 0000	0	
1425	0401 0000	0	
1426	0402 0002	2	/BINARY
1427	0403 0000	0	/BLK NO
1430	0404 0000	0	/NO OF BLKS
1431		/	
1432	0405 0000	RWPARM, 0000	/UNIT
1433	0406 0020	0020	/BUFFER ADDR
1434	0407 0000	0	/BLK NO
1435	0410 0000	0	/NO OF BLKS
1436		/	
1437		LMODE	
1440		/QUESTIONS	
1441	0411 4040		
1442	0412 4040		
1443	0413 4023		
1444	0414 1116		
1445	0415 0714		
1446	0416 0540		
1447	0417 2022		
1448	0420 0503		
1449	0421 1123		
1450	0422 1117		
1451	0423 1640		
1452	0424 0606		
1453	0425 2443		
1454			
1455	0426 4740		
1456	0427 4347		
1457			
1458	0430 4043		
1459	0431 0640		
1460	0432 4011		
1461	0433 1620		
1462	0434 2524		
1463	0435 4017		
1464			
1465	0436 1643		
1466			
1467	0437 4740		
1468	0440 4306		
1469	0441 4040		
1470	0442 0411		
1471	0443 0114		
1472	0444 4025		
1473	0445 1611		
1474	0446 2477		
1475	0447 4031		
1476	0450 5716		
1477	0451 7461		
1478	0452 3400		
1479			
1480			
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1600			

1450	0457	1625	
1450	0460	1502	
1450	0461	0522	
1450			FUNIT NUMBER<2
1451	0462	7462	
1451	0463	4347	
1451			
1452	0464	4043	
1452	0465	0606	
1452	0466	1114	
1452	0467	0540	
1452	0470	1601	
1452	0471	1505	
1452	0472	4040	
1452	0473	7470	
1452	0474	3400	
1452			FFILE NAME <8\2
1453			QUES3, TEXT 2
1454	0475	4043	
1454	0476	0640	
1454	0477	4025	
1454	0500	1611	
1454	0501	2440	
1454	0502	1625	
1454	0503	1502	
1454	0504	0522	
1454			F UNIT NUMBER<2
1455	0505	7462	
1455	0506	4347	
1455			
1456	0507	4043	
1456	0510	0640	
1456	0511	4002	
1456	0512	1413	
1456	0513	4016	
1456	0514	2515	
1456	0515	0205	
1456	0516	2240	
1456	0517	7463	
1456	0520	3400	
1456			F BLK NUMBER <3\2
1457			QUES4, TEXT 2
1460	0521	4043	
1460	0522	0610	
1460	0523	1727	
1460	0524	4015	
1460	0525	0116	
1460	0526	3140	
1460	0527	2024	
1460	0530	2377	
1460			FWOW MANY PTS? <4
1461	0531	7464	
1461	0532	4347	
1461			
1462	0533	4043	
1462	0534	4740	
1462	0535	5064	
1462	0536	5561	
1462	0537	6062	
1462	0540	6440	
1462	0541	0231	
1462	0542	4020	

(4=1024 BY POWERS OF 2)

FREAL OR

FCOMPLEX? R/C<1\Z
QUES5, TEXT Z

F OUTPUT ON

F DIAL UNIT? Y/N<1\Z
QUES6, TEXT Z

F REPLACE? Y/N<1\Z
QUES11, TEXT Z

5

-5

5

1470	0021	2440	
1476	0630	1722	
1476	0631	4004	
1476	0632	1123	
1476	0633	2014	
1476	0634	0131	
1476	0635	7740	
1476	0636	0657	
1476	0637	0474	
1476			FFFT OR DISPLAY? F/D<1
1477	0640	6143	
1477			
1500	0641	4740	
1500	0642	4347	
1500			
1501	0643	4043	
1501	0644	0624	
1501	0645	2201	
1501	0646	1623	
1501	0647	0617	
1501	0650	2215	
1501	0651	4017	
1501			FTRANSFORM OR
1502	0652	2243	
1502			
1503	0653	4740	
1503	0654	4306	
1503	0655	1116	
1503	0656	2605	
1503	0657	2223	
1503	0660	0577	
1503	0661	4024	
1503	0662	5711	
1503	0663	7461	
1503	0664	3400	
1503			FINVERSE? T/I<1\2
1504			QUES13, TEXT 2
1505	0665	4306	
1505	0666	2710	
1505	0667	1103	
1505	0670	1040	
1505	0671	0411	
1505	0672	2320	
1505	0673	1401	
1505	0674	3177	
1505			FWHIGH DISPLAY?<1
1506	0675	7461	
1506	0676	4347	
1506			
1507	0677	4043	
1507	0700	4740	
1507	0701	4040	
1507	0702	4040	
1507	0703	2250	
1507	0704	0501	
1507			R(EAL)
1510	0705	1451	
1510	0706	4347	
1510	0707	4040	
1510	0710	4040	
1510	0711	4011	
1510	0712	5015	
1510	0713	0107	

I (MAGINARY)

M (AGNITUDE)

S (CALE FACTOR)

LINE FEED (RESTART) \Z

/MESSAGES
MSG1, TEXT Z

F CANNOT FIND

1510	0715	0122
1511	0716	3151
1511	0717	4347
1511	0720	4040
1511	0721	4040
1511	0722	4015
1511	0723	5001
1511	0724	0716
1511	0725	1124
1511	0726	2504
1511	0727	0551
1512	0730	4347
1512	0731	4040
1512	0732	4040
1512	0733	4023
1512	0734	5003
1512	0735	0114
1512	0736	0540
1512	0737	0601
1512	0740	0324
1512	0741	1722
1512	0742	5143
1513	0743	4740
1513	0744	4040
1513	0745	4040
1513	0746	1411
1513	0747	1605
1513	0750	4006
1513	0751	0505
1513	0752	0450
1513	0753	2205
1513	0754	2324
1513	0755	0122
1513	0756	2451
1513	0757	3400
1513	0760	4347
1516	0761	4043
1517	0762	0640
1517	0763	4040
1517	0764	4003
1517	0765	0116
1517	0766	1617
1517	0767	2440
1517	0770	0611
1517	0771	1604
1520	0772	4347
1520	0773	4043
1521	0774	4740
1521	0775	4040
1521	0776	4040
1521	77	4040
1521	-000	4040
1521	1001	4010

1521	1002	1124			
1521	1003	4022			
1521	1004	0524			
1521	1005	2522			
1521	1006	1640			
1521	1007	2417			
1521	1010	4003			
1521	1011	1716			
1521	1012	2434			
1521					
1522					
1523	1013	4347			
1523					
1524	1014	4043			
1524	1015	0640			
1524	1016	4040			
1524	1017	4016			
1524	1020	1740			
1524	1021	2320			
1524	1022	0103			
1524					
1525	1023	0543			
1525					
1526	1024	4740			
1526	1025	4347			
1526	1026	4040			
1526	1027	4040			
1526	1030	4040			
1526	1031	4010			
1526	1032	1124			
1526	1033	4022			
1526	1034	0524			
1526	1035	2522			
1526	1036	1640			
1526	1037	2417			
1526	1040	4003			
1526	1041	1716			
1526	1042	2434			
1526					
1527					
1530	1043	0000			
1531					
1532					
1533					

MSG2, TEXT Z HIT RETURN TO CONT\Z

F NO SPACE

/ ANSWER, 0 *.+6

EJECT

HIT RETURN TO CONT\Z

1535	7052	0000	/SQUARE ROOT		
1536	7053	3273	SQRT, 0		
1537	7054	1142	DCA	ROOT	/CLR ROOT
1540	7055	7640	TAD	DPSQ	/IF SQ IS 0, EXIT
1541	7056	5262	SZA CLA		
1542	7057	1143	JMP	NOT0	
1543	7058	7650	TAD	DPSQ+1	
1544	7059	5652	SNA CLA		
1545	7060	1143	JMP I	SQRT	
1546	7061	7110	TAD	DPSQ+1	/1ST APPROX OF ROOT
1547	7062	1140	CLL RAR		/DIVIDE BY 2
1550	7063	1140	TAD	C2000	/+1/2
1551	7064	3273	DCA	ROOT	
1552	7065				
1553	7066	1142	SQRT2,	DPSQ	/SUM OF SQUARES
1554	7067	7421	MQL		
1555	7070	1143	TAD	DPSQ+1	
1556	7071	7100	CLL		
1557	7072	7407	DVI		/DIVIDE BY ROOT APPROX
1560	7073	0000	0		
1561	7074	7430	SZL		
1562	7075	5314	JMP	SQRT1	/OVERFLOW
1563	7076	7200	CLA		
1564	7077	7501	MQA		/QUOTIENT TO AC
1565	7100	1273	TAD	ROOT	/ADD APPROX TO QUOTIENT
1566	7101	7001	IAC		
1567	7102	7110	CLL RAR		/DIVIDE BY 2
1570	7103	3035	DCA	SINE	/SAVE
1571	7104	1273	TAD	ROOT	/SUBTRACT OLD ROOT FROM NEW ONE
1572	7105	7041	CIA		
1573	7106	1035	TAD	SINE	
1574	7107	7650	SNA CLA		
1575	7110	5314	JMP	SQRT1	/=0
1576	7111	1035	TAD	SINE	
1577	7112	3273	DCA	ROOT	/NEW ROOT
1600	7113	5266	JMP	SQRT2	
1601					
1602	7114	1273	SQRT1,	ROOT	
1603	7115	5652	JMP I	SQRT	
1604					
1605					
1606			EJECT		

1607	PHODE		
1610	/MOVE PTS FROM ONE AREA TO ANOTHER		
1611	/10 = OLD BUFFER		
1612	/11 = NEW "		
1613	/IF CMPFLG=1, COMPLEMENT VALUE		
1614	MOVPTS, 0	7116 2000	
1615	CONF1	7117 6211	
1616	NXTPT, TAD	7120 1130	
1617	CLL RAR	7121 7113	
1620	TAD I 10	7122 1410	
1621	SZL	7123 7430	
1622	CIA I 11	7124 7041	
1623	DCA I 11	7125 3411	
1624	ISZ	7126 2034	
1625	JMP	7127 5320	
1626	NXTPT	7130 6201	
1627	CONF1	7131 5716	
1630	JMP I MOVPTS		
1631			
1632	/MOVE REAL OR MAGNITUDE VALUES		
1633	/FROM 0 TO 2000		
1634	/AND SWAP HALVES		
1635	/DO NOT COMPLEMENT		
1636	MVRLMG, 0	7132 0000	
1637		7133 7040	
1640	CMA	7134 3010	
1641	DCA	7135 1037	
1642	TAD	7136 1110	
1643	TAD	7137 3011	
1644	DCA	7140 1033	
1645	TAD	7141 3034	
1646	DCA	7142 3130	
1647	JMS I	7143 4527	
1650	CMA	7144 7040	
1651	TAD	7145 1037	
1652	DCA	7146 3010	
1653	TAD	7147 1110	
1654	DCA	7150 3011	
1655	TAD	7151 1033	
1656	DCA	7152 3034	
1657	JMS I	7153 4527	
1660	JMP I	7154 5732	
1661	MVRLMG		
	EJECT		

			/OLD ADDR OF 1ST 1/2 = 0
			/NEW ADDR OF 1ST 1/2 = 2000 + 1/2 NO OF PTS
			/MOVE 1/2 NO OF PTS
			/DON'T COMPLEMENT
			/MOVE THEM
			/OLD ADDR OF 2ND 1/2 = 1/2 NO OF PTS
			/NEW ADDR OF 2ND 1/2 = 2000
			/1/2 NO OF PTS
			/MOVE THEM

ACDF0 1202
ADDR 41
ADDR 35
ADDW0S 1157
ADD1 1174
ADD2 0033
ADJSGN 0570
ANSWER 7043
ARG2 1020
ASK 4720
ASKX 4740
ASK2 4523
ASK2X 4571
ASK3 4572
ASK3X 4626
ASR 7415
BIGSNU 0012
BOUND 1516
BUFHI 1574
BUFLO 1575
BUFPTR 1504
BUILD 0544
C 0032
CAM 7621
CCDF0 1401
CCIA 0162
CCLR 0146
CDF0 6201
CDF1 6211
CHKEND 4650
CHKHI 1473
CHKPT 0515
CKEND 4211
CMPFLG 0130
CNOP 0163
CNOTS 0677
CONT 1400
CONV 4627
CONVER 4674
CORVAL 0114
COSINE 0036
COUNT 0116
CSAM 1351
CURCNT 1573
CURDIS 1551
CURL0P 1563
CURRTN 1450
CURSAM 0101
C1000 0137
C1777 0110
C2000 0140
C6000 0101
DADD 1504
DATTAP 4061
DBLHI 1541
DBLLO 1547
DISFLG 4242
DISPER 6051
DISPLY 6001
DISPL1 6022

DOIFFT 0047
DPI MAG 6055
DPMAG 6270
DPREAL 6117
DPSCAL 6226
DPSJ 0142
DSCLOC 1244
DSCLOP 1275
DSCWD 1261
DVI 7407
ENDHI 1577
ENDLO 1576
ERRCHK 4660
F 0007
FDV 6375
FDV2RW 4466
FFT 0400
FIF 4251
FLIP 1046
FLIPCT 0063
FREE 1365
FRESAM 0105
FT 4365
GETRIG 0045
GI 0040
GQ1000 6147
GR 0037
GRET10 0073
GR9 6244
IDORA 1200
IFCOM 4203
IFDIAL 4026
IFDISP 4265
IFFFT 4231
IFFT 0147
IFI 4261
IFT 4363
IFTFLG 4356
IMPTR 6336
INDEX 1134
INFILE 4555
INVERT 0043
INVRT 1040
K 0031
KBUFHI 1241
KBUFLO 1242
KIDORA 0117
KMNADR 1236
KMNFLD 1235
KMXADR 1240
KMXFLD 1237
KRDORA 0120
KYSICAL 0136
L 0005
LARGER 6216
LDF4 0144
LEFTX 1363
LESS10 0074
LOADDR 61
LOOP1 04
LSR 741
L

MAXFLD 1474
MAXNU 21
MINADR 16
MINFLD 1415
MINPTS 0131
MNOVR2 0022
MOVVIS 1561
MOVINP 4063
MOVPTS 7116
MPLIER 4701
MQA 7501
MQL 7421
MSG1 6760
MSG2 7013
MULPLY 4676
MULT 0044
MULTIP 1000
MUY 7405
MVRLMG 7132
M1 0070
M1K 0141
M10 0072
M1000 1527
M11 0077
M12 0071
M215 0102
M261 0106
M321 0103
M340 0105
M353 0104
M4 0075
M400 0107
M4000 0067
M5 0100
M70 1476
N 0003
NMI 7411
NOROT 0564
NOSPAC 4423
NOSWPI 6110
NOSWPR 6124
NOTNOR 1172
NOT0 7062
NOVER4 0020
NOWSTR 4377
NO4MIK 1133
NU 0004
NUM8KS 4503
NUM8KX 4522
NXTCHR 4634
NXTDF 1466
NXTMAG 6320
NXTPNT 1441
NXTPT 7120
OCTL 4711
OF12BU 4000
OKEND 1463
OKFLO 1407
ONDAT 4452
OUTQES 4273
OUTUNT 431W

POPSCL 6054
PDSPER 6372
PFDV7 0123
PI 0026
PIFTFG 0133
PLEFTX 0125
PMRLMG 0126
PMVDIS 0124
PMVPTS 0127
PNSWPR 6374
PPREPR 6371
PR 0025
PREAD 0134
PRELFG 0132
PREPAR 6130
PRFRSH 0122
PROC 4357
PSHOWT 0121
PSQRT 6373
PTS 4064
PWRITE 0135
P401 1243
Q 0027
GAINIT 1000
QARFSH 1053
QI 0024
QR 0023
QUAD1 1110
QUAD2 1072
QUESNO 4734
QUES1 6411
QUES11 6625
QUES13 6665
QUES2 6453
QUES3 6475
QUES4 6521
QUES5 6571
QUES6 6612
RBUILD 0064
RDATA 4320
RDORA 1343
REALFG 6053
RECHK 0066
REDPLY 6160
RELPTR 6324
REPL 4446
RESEIC 0065
REVERS 0705
RFRSH 6171
ROOT 7073
ROTAT 4121
ROT1 4136
RTNCDF 1257
RWPARM 6405
S 0006
SAMNAM 4430
SCA 7441
SCAL 0053
SCALE 0340
SCL 74
SCLFAC 64
SCR4 0140

SETC 0542
SETDF 541
SETFL 32
SGNADJ 0161
SHCHK 0255
SHFLAG 2054
SHFT1 1277
SHFT2 1114
SHFT3 1125
SHIFCT 2562
SHIFT1 0056
SHIFT2 0057
SHIFT3 0060
SHL 7413
SHOSQL 6247
SHOWIT 6162
SIGN 1037
SINE 0035
SINLOC 0050
SINRET 1122
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